

Linear Algebra Review Lecture

ESE 551 Linear Dynamic Systems

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August 26, 2024

Overview

This is a review lecture on linear algebra and some fundamental results in functional analysis. We assume that you are already familiar with vectors, matrices, and other basic concepts in linear algebra.

Suggested readings:

- *Linear Algebra and Its Applications* by Gilbert Strang, 4th edition, Academic Press, 2009.
- *Introduction to Matrix Analysis*, by Richard Bellman, McGraw-Hill, 1960.
- *Theory of Matrices*, by Felix R. Gantmacher, vol. I & II, Chelsea, 1960.
- *Linear Algebra*, by Kenneth M. Hoffman and Ray A. Kunze, Pearson, 1971.
- *Linear Algebra and Its Applications*, by Peter D. Lax, Wiley, 2007.
- *Linear Algebra, 4th Edition*, by Stephen H. Friedberg, Prentice-Hall Of India Pvt. Limited, 2003.

Vector Spaces

- Vector space
- Linear subspace / Linear variety
- Span / Linear combination / Linear Independence
- Basis / Dimension

Vector Spaces

Why **Vector Spaces**?

- Many concepts related to vectors in Euclidean space can be extended to other mathematical systems $\Rightarrow$ think of a vector space as a collection of objects that behave similarly to vectors in $\mathbb{R}^n$.

Vector Spaces

Definition 1 (Vector Space / Linear Space).

A **vector space** V over a **field** $\mathbb{F}$ is a set of elements called **vectors**, together with two operations, *vector addition*: $+: V \times V \rightarrow V$, defined by $x, y \in V \mapsto x + y \in V$,
scalar multiplication: $\cdot: \mathbb{F} \times V \rightarrow V$, defined by $\alpha \in \mathbb{F}, x \in V \mapsto \alpha \cdot x \in V$,
such that $\forall x, y, z \in V$ and $\forall \alpha, \beta \in \mathbb{F}$, the following properties (*axioms*) hold:

1. $x + y = y + x$ (additive commutativity)
2. $(x + y) + z = x + (y + z)$ (additive associativity)
3. $\exists! 0 \in V$ s.t. $x + 0 = 0 + x = x$ (additive identity)
4. $\exists! (-x) \in V$ s.t. $x + (-x) = 0$ (additive inverse)
5. $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$ (scalar distributivity)
6. $(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$ (vector distributivity)
7. $(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$ (multiplicative associativity)
8. $\exists 1 \in \mathbb{F}$ s.t. $1 \cdot x = x$ (multiplicative identity)

Vector Spaces

Example 2 (Vector Space).

- **The space of n -tuples $\mathbb{F}^n$ over $\mathbb{F}$:** the set of all n -tuples $x = (x_1, \dots, x_n)$ with $x_k \in \mathbb{F}$, $+$ and $\cdot$ defined componentwise
- **The space of infinite sequences $\mathbb{F}^\infty$ over $\mathbb{F}$:** the set of all infinite sequences $x = (x_1, \dots, x_k, \dots) = \{x_k\}_{k \in \mathbb{N}}$ with $x_k \in \mathbb{F}$, $+$ and $\cdot$ defined componentwise
- **The space of $m \times n$ matrices $\mathbb{F}^{m \times n}$ over $\mathbb{F}$:** the set of all $m \times n$ matrices A with $A_{ij} \in \mathbb{F}$, $(A + B)_{ij} = A_{ij} + B_{ij}$, $(\alpha \cdot A)_{ij} = \alpha \cdot A_{ij}$
- **The space of all functions $\mathcal{F}(S, \mathbb{F})$ over $\mathbb{F}$:** let S be any nonempty set, the set of all $f : S \rightarrow \mathbb{F}$ defined by $s \mapsto f(s)$, with $(f + g)(s) = f(s) + g(s)$ and $(c \cdot f)(s) = c \cdot f(s)$

Vector Spaces

Example 3 (Verify the following space is a vector space).

- **The space of polynomial functions** $P(\mathbb{F})$: The set of all functions f from $\mathbb{F}$ to $\mathbb{F}$ of the form $f(x) = c_0 + c_1x + \cdots + c_nx^n$

Vector Spaces

Important examples:

- Often take $\mathbb{F} = \mathbb{R}$ (or $\mathbb{C}$), leading to $V = \mathbb{R}^n, \mathbb{R}^\infty, \mathbb{R}^{m \times n}, \dots$
- **The space of all real-valued continuous functions** $C(\Omega)$:
 $V = C([t_0, t_1], \mathbb{R}) = \{f : [t_0, t_1] \subset \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ continuous}\}$
- **The space of m -tuple real-valued continuous functions** $V = C^m([t_0, t_1], \mathbb{R}^m)$: the space of m -tuples $f : [t_0, t_1] \rightarrow \mathbb{R}^m$ whose elements are continuous functions on $[t_0, t_1]$.
- **Lebesgue spaces** L^p the space of measurable functions whose p -norm is finite, i.e., $(\int_S |f|^p d\mu)^{1/p} < \infty$: the space of square-integrable functions L^2
- **The sequence space** ℓ^p : the space of sequences whose series are absolutely convergent ℓ^1 , the space of square-summable sequences ℓ^2 , the space of bounded sequences ℓ^∞
- Normed vector space, Banach space, Inner product space, Hilbert space, ...

Vector Spaces

Remark:

- Notice that a vector space defines scalar-vector multiplication, but not vector-vector multiplication.
- A *field* is a ring whose nonzero elements form an abelian group under multiplication. This means it is defined with two operations, addition and multiplication, which obey closure, associativity, identity elements, inverse elements, commutativity (the order of elements in an operation doesn't matter).
- In other words, a field is a set $\mathbb{F}$ of numbers with the property that if $a, b \in \mathbb{F}$, then $a + b$, $a - b$, ab and a/b are also in $\mathbb{F}$ (assuming, of course, that $b \neq 0$ in the expression a/b).
e.g. $\mathbb{R}, \mathbb{C}, \mathbb{Q}, \mathbb{N}, \mathbb{Z}$

Vector Spaces

Why **Subspaces**?

- Vector spaces may be formed from subsets of other vectors spaces. These subsets are referred to as "subspaces".
- Many concepts and results in linear algebra are formulated in terms of "subspaces". For example, topics include dimensions, solution sets to linear equations, eigenspaces, spans, orthogonal complements, and the image and preimage are all related to or defined by subspaces.

Vector Spaces

Definition 4 (Subspace / linear subspace).

A nonempty subset S of a vector space V is called a **subspace** of V if $\alpha x + \beta y \in S$ for every $x, y \in S$ and every $\alpha, \beta \in \mathbb{R}$ (**closed under vector addition and scalar multiplication**).

Remark:

- By definition, a subspace must contain the null vector 0 .
- A subspace not equal to the entire space is said to be a *proper subspace*.
- Alternatively, one may see the definition of subspace as follows: A subspace S of a vector space V is a subset of V that satisfies:
 1. $0 \in S$.
 2. $x + y \in S$ for every $x, y \in S$.
 3. $\alpha x \in S$ for every $x \in S$ and every $\alpha \in \mathbb{R}$.

Vector Spaces

Theorem 5.

- In any vector space V , V and $\{0\}$ are subspaces of V .
- The sum of any collection of subspaces is a subspace.
- The intersection of any collection of subspaces is a subspace.
- The union of a collection of subspaces **totally ordered by set inclusion** is a subspace.

Remark:

- The union of a collection of subspaces is **not** a subspace.
- e.g.: Take S_1 and S_2 to be x -axis and y -axis, both subspaces of $\mathbb{R}^2$. Then, $v_1 = (1, 0) \in S_1 \cup S_2$ and $v_2 = (0, 1) \in S_1 \cup S_2$ but $v_1 + v_2 = (1, 1) \notin S_1 \cup S_2$. Hence, the union $S_1 \cup S_2$ is not a subspace of $\mathbb{R}^2$.

Example 6 (exercise).

- The set of all $n \times n$ symmetric matrices is a subspace of $\mathbb{R}^{n \times n}$.
- The set of all $n \times n$ diagonal matrices is a subspace of $\mathbb{R}^{n \times n}$.
- The set of all $n \times n$ matrices having trace equal to 0 is a subspace of $\mathbb{R}^{n \times n}$.
- Let $P_n(\mathbb{F})$ consist of all polynomials in $P(\mathbb{F})$ having degree less than or equal to n , then $P_n(\mathbb{F})$ is a subspace of $P(\mathbb{F})$.
- $C(\mathbb{R})$ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{R})$.
- The set of all functions in $\mathcal{F}(\mathbb{R}, \mathbb{C})$ that have derivatives of all orders C^∞ is a subspace of $\mathcal{F}(\mathbb{R}, \mathbb{C})$.

Vector Spaces

Definition 7 (Linear Variety / linear manifold / affine subspace).

A **linear variety** L in V is a translated subspace $L = x_0 + S = \{x_0 + x : \forall x \in S\}$ for $x_0 \in V$ and where $S \subset V$ is a subspace.

Example 8.

Identify if the following is a linear subspace and/or a linear variety:

- $S_1 = \{x \in \mathbb{R}^3 \mid x_1 = x_2, x_2 = x_3\}$
- $S_2 = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 1\}$

- The solution set of a nonhomogeneous system of linear equations $Ax = b$ can be described in terms of the solution set of the homogeneous system $Ax = 0$ with a solution x_0 to $Ax = b$.

Vector Spaces

Example 9 (exercise on vector space/subspace).

Let W be a subspace of a vector space V over $\mathbb{F}$. For any $v \in V$, the set $\{v\} + W := \{v + w : w \in W\}$ is called *the coset of W containing v* , denoted by $v + W$. Further, define the addition and scalar multiplication in the collection $S = \{v + W : v \in V\}$ of all cosets of W as follows:

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W, \quad \alpha(v_1 + W) = \alpha v_1 + W$$

for all $v_1, v_2 \in V$ and $\alpha \in \mathbb{F}$. Prove that

1. $v + W$ is a subspace of V if and only if $v \in W$.
2. $v_1 + W = v_2 + W$ if and only if $v_1 - v_2 \in W$.
3. The preceding operations are well defined.
4. The set S is a vector space with the operations defined above. This vector space is called the **quotient space of V modulo W** and is denoted by V/W .

Vector Spaces

Example 10.

Let W_1 be the set of all skew-symmetric $n \times n$ matrices with entries from $\mathbb{R}$ and W_2 be the set of all symmetric $n \times n$ matrices with entries from $\mathbb{R}$. Show that

- Both W_1 and W_2 are subspaces of $\mathbb{R}^{n \times n}$.
- $\mathbb{R}^{n \times n} = W_1 \oplus W_2$.

Vector Spaces

Why **Basis**?

- A basis of a vector space is significant because it provides a fundamental framework for representing and working with vectors in that space. For instance, it ensures the uniqueness of vector representation, defines the dimension of the vector space, and assigns coordinates to vectors for easier manipulation and understanding...

Vector Spaces

Definition 11 (Span).

Let $S = \{x_1, \dots, x_m\} \subset V$ be a nonempty subset of a vector space V . The **span** of S , denoted by $\text{span}(S)$, is the set consisting of all **linear combinations** of the vectors in S .

That is,

$$\text{span}(S) = \text{span}\{x_1, \dots, x_m\} = \left\{ v \in V : v = \sum_{i=1}^m \alpha_i x_i \text{ for some } \alpha_i \in \mathbb{R} \right\}$$

Definition 12 (Linear Independence).

A set of vectors $\{x_1, \dots, x_m\}$ in a vector space V is said to be **linearly independent** if $\sum_{i=1}^m \alpha_i x_i = 0$ with $\alpha_1, \dots, \alpha_m$ being scalars implies that $\alpha_i = 0$ for all $i = 1, \dots, m$. That is,

$$\sum_{i=1}^m \alpha_i x_i = 0 \Rightarrow \alpha_i = 0 \quad \forall i = 1, \dots, m$$

An infinite set of vectors is linearly independent if every finite subset is linearly independent.

Vector Spaces

Remark:

- A subset S of a vector space V **generates** (or **spans**) V if $\text{span}(S) = V$. In this case, we also say that the vectors of S generate (or span) V .

Definition 13 (Basis).

A **basis** $\mathcal{B}$ for a vector space V is a linearly independent subset of V that generates V . If $\mathcal{B}$ is a basis for V , we also say that the vectors of $\mathcal{B}$ form a basis for V .

- Let V be finite-dimensional. If $\mathcal{B}$ for V is endowed with a specific order, then we say $\mathcal{B}$ is an **ordered basis** for V .

Theorem 14.

$\mathcal{B}$ is a basis for V if and only if each element in V can be uniquely expressed as a linear combination of vectors of $\mathcal{B}$.

Vector Spaces

Definition 15 (Dimension).

The **dimension** of V , denoted by $\dim(V)$, is the number of vectors in the basis set $\mathcal{B}$. The space V is **finite-dimensional** if it has a finite basis, i.e., $\dim(V) < \infty$.

Lemma 16.

Let V be a vector space.

- If V has a finite basis, then every basis for V contains the same number of vectors.*
- If V is infinite-dimensional, then every basis for V has the same cardinality.*

Vector Spaces

Example 17.

- In $\mathbb{R}^n$, $\mathcal{B} = \{e_1, \dots, e_n\}$ is the "standard ordered" basis for $\mathbb{R}^n$, where e_i denotes the vector whose i th entry is 1 and 0 otherwise.
- In $\mathbb{R}^{m \times n}$, $\mathcal{B} = \{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis for $\mathbb{R}^{m \times n}$, where E_{ij} denotes the matrix whose only nonzero entry is 1, in the i th row and j th column.
- In $P_n(\mathbb{F})$ (the space of all polynomials with degree less than or equal to n), $\mathcal{B} = \{1, x, x^2, \dots, x^n\}$ is the standard ordered basis for $P_n(\mathbb{F})$.
- In $P(\mathbb{F})$ (the space of all polynomials), $\mathcal{B} = \{1, x, x^2, \dots\}$ is a basis.

Remark:

- A linear combination by definition only involves "finitely" many vectors.
- A basis $\mathcal{B}$ can indeed have an infinite number of elements. However the span of $\mathcal{B}$, is the set of vectors that are written as "finite" linear combinations of elements of $\mathcal{B}$.

Vector Spaces

Example 18.

Let W be a subspace of a finite-dimensional vector space V , and consider the basis $\{u_1, u_2, \dots, u_k\}$ for W . Let $\{u_1, u_2, \dots, u_k, u_{k+1}, \dots, u_n\}$ be an extension of this basis to a basis for V . Show that

1. $\{u_{k+1} + W, u_{k+2} + W, \dots, u_n + W\}$ is a basis for V/W .
2. $\dim(V/W) = \dim(V) - \dim(W)$.

Linear Transformations

- Linear transformation (linear mapping)
- Null space / Range space / Dimension theorem
- Matrix representation
- Invertibility
- Isomorphism / Homomorphism / Endomorphism / Automorphism
- Dual space

Linear Transformations

Why **Linear transformations**?

- Linear transformations are functions defined on vector spaces that "preserve their structure". For example, in calculus, differentiation and integration are key examples of linear transformations, which allow the reformulation of differential and integral equations in terms of vector spaces. In geometry, rotations, reflections, and projections also exemplify linear transformations and are used to study rigid motions.

Linear Transformations

Definition 19 (Mapping / Map / Transformation).

A **mapping** $L : V \rightarrow W$ is a rule ("function") that assigns every element $x \in V$ to an element $z \in W$, denoted by $L(x) = z$, where V and W are the *domain* and the *codomain* of the mapping L . Additionally, $L(x)$ is called the *image* of x (under L), and x is called a *preimage* of $L(x)$ (under L).

Definition 20 (Linear mapping / linear transformation).

Let V and W be vector spaces over $\mathbb{F}$. A mapping $L : V \rightarrow W$ is **linear** if $\forall x, y \in V$ and $\forall \alpha \in \mathbb{F}$, we have

1. $L(x + y) = L(x) + L(y)$ (Additivity)
2. $L(\alpha x) = \alpha L(x)$ (Homogeneity)

In this case, we call $L : V \rightarrow W$ is a **linear transformation from V to W** .

Linear Transformations

Example 21.

- The **differentiation transformation**: Let $L : P_n(\mathbb{R}) \rightarrow P_{n-1}(\mathbb{R})$ be defined by $L(f(x)) = Df(x)$, where $Df(x)$ denotes the derivative of $f(x)$.
- Let $\Omega \subset \mathbb{R}$ and let $L : C(\mathbb{R}) \rightarrow \mathbb{R}$ be defined by $L(f) = \int_{\Omega} f(x)dx$ for all $f \in C(\mathbb{R})$.
- The **identity transformation** $\text{id}_V : V \rightarrow V$ defined by $\text{id}_V(x) = x$ for all $x \in V$.
- The **zero transformation** $T_0 : V \rightarrow W$ defined by $T_0(x) = 0$ for all $x \in V$.
- The **left-multiplication transformation** $L_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ defined by $L_A(x) = Ax$ for $x \in \mathbb{R}^n$, where $A \in \mathbb{R}^{m \times n}$.

Linear Transformations

Remark:

- **Homomorphism:** In linear algebra, a linear map is also referred to as a homomorphism of vector spaces.
- The set of linear mappings between two vector spaces V and W is itself a vector space. It is usually denoted by $\mathcal{L}(V, W)$ or $\text{Hom}(V, W)$.

Linear operator: We sometimes use *linear operator* interchangeably with *linear transformation*, especially in the content of functional analysis. In some linear algebra book, one might find *linear operator* specifically refers to a linear transformation from a vector space to itself, i.e., $T : V \rightarrow V$. This is why we commonly hear phrases like "Let T be a linear operator on a separable Hilbert space" without specifying the codomain.

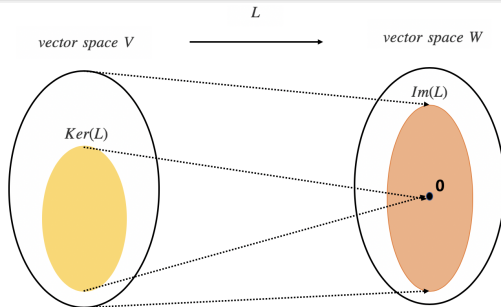
Linear Transformations

Definition 22 (Null space & Range space).

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear.

We define the **null space** (or **kernel**) $\mathcal{N}(L)$ of L to be the set of all vectors x in V such that $L(x) = 0$; that is, $\mathcal{N}(L) = \text{Ker}(L) = \{x \in V : L(x) = 0\}$.

We define the **range** (or **image**) $\mathcal{R}(L)$ of L to be the subset of W consisting of all images (under L) of vectors in V ; that is, $\mathcal{R}(L) = \text{Im}(L) = \{y \in W : y = L(x), \forall x \in V\}$.



Linear Transformations

Example 23.

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear. Show that

- $\mathcal{N}(L)$ and $\mathcal{R}(L)$ are subspaces of V and W , respectively.
- If $\mathcal{B} = \{v_i\}_{i=1}^n$ is a basis for V , then

$$\mathcal{R}(L) = \text{span}(L(\mathcal{B})) = \text{span}(\{T(v_1), T(v_2), \dots, T(v_n)\})$$

Remark:

- This provide a way for finding a spanning set for the range of a linear transformation and thus a way of **constructing basis for the range of a linear transformation**.
- The same idea can be used for the case when $\mathcal{B}$ is infinite, that is,
 $\mathcal{R}(L) = \text{span}(\{T(v) : v \in \mathcal{B}\})$.

Linear Transformations

Definition 24 (Nullity & Rank).

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear.

If $\mathcal{N}(L)$ and $\mathcal{R}(L)$ are finite-dimensional, then we define the **nullity** of L , denoted $\text{nullity}(L)$, and the **rank** of L , denoted $\text{rank}(L)$, to be the dimensions of $\mathcal{N}(L)$ and $\mathcal{R}(L)$, respectively.

- $\text{nullity}(L) = \dim(\mathcal{N}(L))$, $\text{rank}(L) = \dim(\mathcal{R}(L))$

Theorem 25 (Dimension theorem / Rank-nullity theorem).

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear. If V is finite-dimensional, then

$$\text{nullity}(L) + \text{rank}(L) = \dim(V)$$

- $\dim(\mathcal{N}(L)) + \dim(\mathcal{R}(L)) = \dim(V)$.

Linear Transformations

Remark:

- The preimage of an element in the range $\mathcal{R}(f)$ need not be unique.
e.g. $f(x) = x^2 + 1$, $f^{-1}(10) = 3$ or -3 .
- A function $f : A \rightarrow B$ is **one-to-one/injective** which maps distinct elements of its domain to distinct elements. In other words, f is one-to-one if each element of the range has a unique preimage, i.e., $f(x) = f(y)$ implies $x = y$.
- A function $f : A \rightarrow B$ is **onto/surjective** if for every element $y \in B$, there exists at least one element $x \in A$ such that $f(x) = y$. In other words, f is onto if and only if the range of f equals the codomain of f .
- A function is **bijective** if it is both injective and surjective. Such function is called a bijective function, or an **one-to-one correspondence**.

Linear Transformations

Theorem 26.

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear. Then L is one-to-one if and only if $\mathcal{N}(L) = \{0\}$.

Proof:

Suppose that L is one-to-one and $x \in \mathcal{N}(L)$. Then $L(x) = 0 = L(0)$. Since L is one-to-one, we have $x = 0$. Hence $\mathcal{N}(L) = \{0\}$. Now assume that $\mathcal{N}(L) = \{0\}$ and suppose that $L(x) = L(y)$. Then $0 = L(x) - L(y) = L(x - y)$. Therefore $x - y \in \mathcal{N}(L) = \{0\}$. So $x - y = 0$, or $x = y$. This means that L is one-to-one.

Linear Transformations

Example 27.

Let $L : V = P_2(\mathbb{R}) \rightarrow W = P_3(\mathbb{R})$ be the linear transformation defined by

$$L(f(x)) = 2f'(x) + \int_0^x 3f(t)dt.$$

Show that if L is one-to-one or onto or both or none.

Linear Transformations

Theorem 28.

Let V and W be vector spaces of finite and equal dimension and let $L : V \rightarrow W$ be linear. Then the following are equivalent:

$$L \text{ is one-to-one} \Leftrightarrow L \text{ is onto} \Leftrightarrow \text{rank}(L) = \dim(V)$$

- If V is not finite-dimensional and $L : V \rightarrow V$ is linear, then it does **not** follow that one-to-one and onto are equivalent (check Example 29, 30 in the next page).

Linear Transformations

Example 29 (exercise).

- Let $L : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be defined by $L(f(x)) = \int_0^x f(t)dt$. Show that L is linear and one-to-one, but not onto.
- Let $L : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be defined by $L(f(x)) = f'(x)$. Show that L is linear and onto, but not one-to-one.

Linear Transformations

Example 30.

Let V be the vector space of infinite sequences. Define the functions $T, U : V \rightarrow V$ by

$$T(a_1, a_2, \dots) = (a_2, a_3, \dots), \quad U(a_1, a_2, \dots) = (0, a_1, a_2, \dots).$$

T and U are called the **left shift** and **right shift** operators on V , respectively. Prove that

1. T and U are linear.
2. T is onto, but not one-to-one.
3. U is one-to-one, but not onto.

Linear Transformations

Example 31.

Let V be a vector space and W be a subspace of V . Define the mapping $\eta : V \rightarrow V/W$ by $\eta(v) = v + W$ for $v \in V$.

1. Prove that η is a linear and onto and that $\mathcal{N}(\eta) = W$.
2. Suppose that V is finite-dimensional. Use (a) and the Dimension Theorem to derive a formula relating $\dim(V)$, $\dim(W)$, and $\dim(V/W)$.

Linear Transformations

Why **Linear transformations** to **Matrix**:

- Establishing a one-to-one correspondence between matrices and linear transformations in a finite-dimensional vector space setting, we utilize properties of one to study properties of the other.

Linear Transformations

Definition 32 (Matrix representation of a linear map).

Let V and W be two vector spaces and let $\mathcal{B}_V = \{u_1, \dots, u_n\}$ and $\mathcal{B}_W = \{v_1, \dots, v_m\}$ be bases of V and W , respectively. Let $L : V \rightarrow W$ be linear. Then for each j , $1 \leq j \leq n$, there exist scalars a_{ij} , $1 \leq i \leq m$, such that

$$L(u_j) = \sum_{i=1}^m a_{ij} v_i.$$

Using the notation above, we call the $m \times n$ matrix A defined by $[A]_{ij} = a_{ij}$ the **matrix representation of L in the bases $\mathcal{B}_V$ and $\mathcal{B}_W$** .

- The matrix A depends on the bases used for V and W .

Linear Transformations

Example 33.

Let $L : V = P_3(\mathbb{R}) \rightarrow W = P_2(\mathbb{R})$ be defined by $L(f(x)) = f'(x)$. Find the matrix representation of L with respect to standard basis.

Definition 34 (Projection).

Let V be a vector space and W_1 and W_2 be subspaces of V such that $V = W_1 \oplus W_2$. A linear operator $L : V \rightarrow V$ is called the **projection on W_1 along W_2** (or simply projection) if, for $x = x_1 + x_2$ with $x_i \in W_i$, we have $L(x) = x_1$.

- L is a projection if and only if $L = L^2$.

Linear Transformations

Example 35.

Let V be a vector space and let $T : V \rightarrow V$ be linear. Prove that $T^2 = T_0$ if and only if $\mathcal{R}(T) \subset \mathcal{N}(T)$.

Linear Transformations

Example 36.

Let V be a vector space. Determine all linear transformations $T : V \rightarrow V$ such that $T = T^2$. Hint: Note that $x = T(x) + (x - T(x))$ for every $x \in V$, and show that $V = \{y : T(y) = y\} \oplus \mathcal{N}(T)$.

Linear Transformations

Why **Inverse of Linear transformations**?

- Many intrinsic properties of functions are shared by their inverses. For example, in calculus we learn that the properties of being continuous or differentiable are generally retained by the inverse functions.
- We see in this section that the inverse of a linear transformation is also linear. This result greatly aids us in the study of inverses of matrices.

Linear Transformations

Definition 37 (Inverse & Invertible).

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear. A function $U : W \rightarrow V$ is said to be an **inverse** of L if $LU = \text{id}_W$ and $UL = \text{id}_V$. If L has an inverse, then L is said to be **invertible**.

- $LU = L \circ U$ denotes the composition of linear transformations L and U .
- If L is invertible, then the inverse of L is unique and is denoted by L^{-1} .
- $(LU)^{-1} = U^{-1}L^{-1}$.
- Let $L : V \rightarrow W$ be a linear transformation, where V and W are finite-dimensional spaces of equal dimension. Then L is invertible if and only if it is both one-to-one and onto.

Theorem 38.

Let V and W be vector spaces and let $L : V \rightarrow W$ be linear and invertible. Then $L^{-1} : W \rightarrow V$ is linear.

Linear Transformations

- The concept of invertibility helps formalize the observation that some vector spaces are very similar except in the form of their vectors. For example, if we associate each matrix in $\mathbb{R}^{2 \times 2}$ to the 4-tuple $(a, b, c, d) \in \mathbb{R}^4$, we see that addition and scalar multiplication behave the same way in both spaces. That is, in terms of the vector space **structure**, these two vector spaces may be considered identical or "isomorphic".

Definition 39 (Isomorphic & Isomorphism).

Let V and W be vector spaces (over the same field). We say that V and W are **isomorphic** (or V is **isomorphic** to W), denoted by $V \cong W$, if there exists a linear transformation $L : V \rightarrow W$ that is invertible/one-to-one correspondence. Such a linear transformation is called an **isomorphism** from V onto W .

Linear Transformations

Recall:

- A **homomorphism** of vector spaces is a linear map from V to W . The set of all homomorphisms from V to W is denoted by $\mathcal{L}(V, W)$.
- A **isomorphism** of vector spaces is a homomorphism from V to W that is also invertible.

Definition 40 (Endomorphism & Automorphism).

- An **endomorphism** of a vector space V is a vector space homomorphism that maps V to itself. The set of all endomorphisms of V is denoted by $\text{End}(V) = \mathcal{L}(V, V)$.
- An **automorphism** of a vector space V is a vector space isomorphism that maps V to itself. The set of all automorphisms of V is denoted by $\text{Aut}(V)$.

$$\begin{array}{ccc} \textit{Automorphism} & \Rightarrow & \textit{Isomorphism} \\ \Downarrow & & \Downarrow \\ \textit{Endomorphism} & \Rightarrow & \textit{Homomorphism} \end{array}$$

Linear Transformations

Example 41.

If V is a vector space over $\mathbb{F}$, show that $\mathcal{L}(V, V)$ endowed with the scalar multiplication and vector addition defined below is a linear transformation on V . Suppose $T_1, T_2 \in \mathcal{L}(V, V)$ and $\lambda \in \mathbb{F}$, then define

- $T_1 + T_2$ to be the mapping whose value at $v \in V$ is $T_1(v) + T_2(v)$.
 - λT_1 to be the mapping whose value at $v \in V$ is $\lambda(T_1(v))$.
-
- $\text{End}(V) = \mathcal{L}(V, V)$ endowed with the scalar multiplication and vector addition defined above is a vector space over $\mathbb{F}$. Moreover, $\text{Aut}(V)$ is a subspace of $\text{End}(V)$.

Linear Transformations

Example 42.

Let B be an $n \times n$ invertible matrix. Define $\Phi : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ by $\Phi(A) = B^{-1}AB$. Prove that Φ is an isomorphism.

Linear Transformations

Example 43.

Let V denote the vector space of infinite sequences and let $W = P(\mathbb{F})$. Define $T : V \rightarrow W$ by $T(\sigma) = \sum_{i=0}^n \sigma_i x^i$, where n is the largest integer such that $\sigma_n \neq 0$. Prove that T is an isomorphism.

Theorem 44.

Let V and W be finite-dimensional vector spaces (over the same field). Then V is isomorphic to W if and only if $\dim(V) = \dim(W)$.

- If V and W are isomorphic, then either both of V and W are finite-dimensional or both are infinite-dimensional.

Linear Transformations

- As a vector space, the collection of all linear transformations between two given vector spaces may be identified with the appropriate vector space of $m \times n$ matrices.

Theorem 45.

Let V and W be finite-dimensional vector spaces over $\mathbb{R}$ of dimensions n and m , respectively. Then the function $\phi : \mathcal{L}(V, W) \rightarrow \mathbb{R}^{m \times n}$ defined by $\phi(L) = A_L$ (matrix representation of L) for $L \in \mathcal{L}(V, W)$, is an isomorphism.

Linear Transformations

Definition 46 (Dual Space).

For a vector space V over $\mathbb{F}$, we define the **dual space** (or dual vector space) of V to be the vector space $\mathcal{L}(V, \mathbb{F})$, denoted by V^* .

- V^* is the vector space consisting of all *linear functionals* on V .
- If V is finite-dimensional, V and V^* are isomorphic (Theorem 44, 45).

Definition 47 (Dual basis).

Let V be a finite-dimensional vector spaces with basis $\mathcal{B} = \{v_1, \dots, v_n\}$. Then we call the basis $\mathcal{B}^* = \{f_1, \dots, f_n\}$ of V^* that satisfies $f_i(v_j) = \delta_{ij}$, $1 \leq i, j \leq n$ the **dual basis** of $\mathcal{B}$.

Linear Transformations

Why **double dual space**?

- Any finite-dimensional vector space V can be identified in a natural way with its double dual V^{**} . There is, in fact, an isomorphism between V and V^{**} that does not depend on any choice of bases for the two vector spaces.

Linear Transformations

Lemma 48.

Let V^ be the dual space of a vector space V . For a vector $x \in V$, we define $\hat{x} : V^* \rightarrow \mathbb{F}$ by $\hat{x}(f) = f(x)$ for every $f \in V^*$. Then $\hat{x}$ is a linear functional on V^* , so $\hat{x} \in V^{**}$.*

- The correspondence $x \mapsto \hat{x}$ allows us to define the isomorphism between V and V^{**} .

Theorem 49.

*Let V be a finite-dimensional vector space, and define $\psi : V \rightarrow V^{**}$ by $\psi(x) = \hat{x}$. Then ψ is an isomorphism.*

- Although many of the ideas can be extended to the case where V is not finite-dimensional, only a finite-dimensional vector space is isomorphic to its double dual via the map $x \mapsto \hat{x}$. In fact, for infinite-dimensional vector spaces, no two of V , V^* , and V^{**} are isomorphic.

Matrix Operations and Diagonalization

- Matrix inverse and determinant
- Matrix inverse formula
- Eigenvalues and Eigenvectors
- Matrix diagonalization
- Jordan form
- Matrix exponential

Matrix Operations and Diagonalization

Definition 50 (Matrix determinant).

If A is an $n \times n$ matrix with $[A]_{ij} = a_{ij}$, then the **cofactor expansion** of A can be computed by

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij}, \quad C_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij}),$$

where $\tilde{A}_{ij}$ is the $(n-1) \times (n-1)$ (sub)matrix obtained by deleting row i and column j from A . $\det(\tilde{A}_{ij})$ is often referred to as the (i,j) -**minor**, and the scalar $C_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$ is called the (i,j) **cofactor** of A .

Definition 51 (Adjugate matrix / Classical adjoint matrix).

If A is an $n \times n$ matrix with $\det(A) \neq 0$, then A is invertible with, $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$, where $\text{adj}(A) = C^T$ is the **adjugate matrix of A** , i.e., the transpose of the cofactor matrix of A , with $[C]_{ij} = C_{ij}$ as defined above.

Matrix Operations and Diagonalization

Example 52.

Compute $\det(A - tI)$ for $A \in \mathbb{R}^{n \times n}$ of the form

$$A = \begin{pmatrix} 0 & 0 & 0 & & 0 & -a_0 \\ 1 & 0 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & 0 & & 0 & a_2 \\ & \vdots & \ddots & \ddots & & \vdots \\ 0 & 0 & 0 & \ddots & 0 & -a_{n-2} \\ 0 & 0 & 0 & \cdots & 1 & -a_{n-1} \end{pmatrix}$$

Matrix Operations and Diagonalization

Definition 53 (Singular matrix).

An $n \times n$ matrix A is called **singular** if $\det(A) = 0$. It is **nonsingular** or **invertible** if $\det(A) \neq 0$. If A and B are $n \times n$ invertible matrices, then $(AB)^{-1} = B^{-1}A^{-1}$.

Theorem 54.

Let A be an $n \times n$ matrix. The following are equivalent:

1. *A is nonsingular.*
2. *A^T is nonsingular.*
3. *For every nonzero $x \in \mathbb{R}^n$, we have $Ax \neq 0$.*
4. *For every $y \in \mathbb{R}^n$, there exists a unique $x \in \mathbb{R}^n$ such that $Ax = y$.*
5. *There exists a unique $n \times n$ matrix B such that $AB = I = BA$, denoted by A^{-1} .*
6. *The columns of A are linear independent.*
7. *The rows of A are linear independent.*

Matrix Operations and Diagonalization

Theorem 55 (Woodbury matrix identity / Woodbury formula).

Let A and B be square matrices and let C be a matrix of proper dimension. Then we have

$$(A + CBC^{\top})^{-1} = A^{-1} - A^{-1}C(B^{-1} + C^{\top}A^{-1}C)^{-1}C^{\top}A^{-1},$$

provided all the inverse appearing above exist.

Theorem 56 (The Schur complement and the inverse of a block matrix).

Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$. It holds

$$M^{-1} = \begin{bmatrix} Q & -QBD^{-1} \\ -D^{-1}CQ & D^{-1} + D^{-1}CQBD^{-1} \end{bmatrix}$$

where $Q^{-1} = M/D := A - BD^{-1}C$ is called the Schur complement of the block D of the matrix M , provided all the inverse appearing above exist.

Matrix Operations and Diagonalization

Why **Diagonalizable**?

- Computations involving diagonal matrices are generally much simpler. For instance, exponential map, determinant, inverse, ...
- Provide a good coordinate system to use for understanding a linear map (a set of axes built from eigenvectors). For instance, the spectral theorem, PCA, ...

Matrix Operations and Diagonalization

Definition 57 (Eigenvalues & Eigenvectors).

Let T be a linear operator on a vector space V . A nonzero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in \mathbb{F}$ such that $T(v) = \lambda v$. The scalar λ is called the **eigenvalue** corresponding to the eigenvector v .

- Taking $T = L_A$ (the left-multiplication transformation) defined on $\mathbb{R}^{n \times n}$ leads to the definition of matrix eigenvalues and eigenvectors as follows:
- Given a square matrix $A \in \mathbb{R}^{n \times n}$, we say $\lambda \in \mathbb{C}$ is an **eigenvalue** of A and $v \in \mathbb{R}^n$ is the corresponding **eigenvector** if $Av = \lambda v$ for some $v \neq 0$. We denote the set of all eigenvalues of A by $\lambda(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$.

Matrix Operations and Diagonalization

Theorem 58.

Let T be a linear operator on a vector space V and let λ be an eigenvalue of T . A vector $v \in V$ is an eigenvector of T corresponding to λ if and only if $v \neq 0$ and $v \in \mathcal{N}(T - \lambda I)$.

- Given a square matrix $A \in \mathbb{R}^{n \times n}$, $\lambda \in \mathbb{C}$ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$.

Matrix Operations and Diagonalization

Example 59.

Let $C^\infty(\mathbb{R})$ denote the set of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ having derivatives of all orders. (Thus $C^\infty(\mathbb{R})$ includes the polynomial functions, the sine and cosine functions, the exponential functions, etc.) Clearly, $C^\infty(\mathbb{R})$ is a subspace of the vector space $\mathcal{F}(\mathbb{R}, \mathbb{R})$ of all functions from $\mathbb{R}$ to $\mathbb{R}$. Let $T : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ be the function defined by $T(f) = f'$, the derivative of f . (It is easy to verify that T is a linear operator on $C^\infty(\mathbb{R})$.) Determine the eigenvalues and eigenvectors of T .

Matrix Operations and Diagonalization

Definition 60 (Characteristic polynomial).

Given a square matrix $A \in \mathbb{R}^{n \times n}$ with $\lambda(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$, $f(t) = \det(A - tI)$ is a polynomial function of degree n , called the **characteristic polynomial of A** , given by $f(t) = \det(A - tI) = (\lambda_1 - t)(\lambda_2 - t) \cdots (\lambda_n - t)$.

- We say a polynomial $f(t)$ in $P(\mathbb{F})$ **splits** (over $\mathbb{F}$) if there are scalars $c, a_1, a_2, \dots, a_n$ (not necessarily distinct) in $\mathbb{F}$ such that $f(t) = c(t - a_1)(t - a_2) \cdots (t - a_n)$.

Definition 61 (Characteristic polynomial of a linear operator).

Let T be a linear operator on an n -dimensional vector space V with ordered basis $\mathcal{B}$. We define the **characteristic polynomial $f(t)$ of T** to be the characteristic polynomial of A , where A is the matrix representation of T . That is, $f(t) = \det(A - tI)$.

Matrix Operations and Diagonalization

Lemma 62.

Show that similar matrices have the same characteristic polynomial.

- This lemma implies that the definition is independent of the choice of basis $\mathcal{B}$. Thus if T is a linear operator on a finite-dimensional vector space V and $\mathcal{B}$ is an ordered basis for V , then λ is an eigenvalue of T if and only if λ is an eigenvalue of A . For this reason, we often denote the characteristic polynomial of an operator T by $\det(T - tI)$.
- We can reduce the problem of finding the eigenvectors of a linear operator on a finite-dimensional vector space to the problem of finding the eigenvectors of a matrix.

Matrix Operations and Diagonalization

Example 63.

Let T be the linear operator on $P_2(\mathbb{R})$ defined by $T(f(x)) = f(x) + (x+1)f'(x)$ and let $\mathcal{B}$ be the standard ordered basis for $P_2(\mathbb{R})$. Find the eigenvalue of T .

Matrix Operations and Diagonalization

Definition 64 (Eigenspaces).

Let T be a linear operator on a vector space V , and let λ be an eigenvalue of T . Define $E_\lambda = \{v \in V : T(v) = \lambda v\} = \text{Ker}(T - \lambda I_V)$. The set E_λ is called the **eigenspace** of T corresponding to the eigenvalue λ .

- Analogously, we define the eigenspace of a square matrix A to be the eigenspace of L_A .
- Given a particular eigenvalue λ of the $n \times n$ matrix A , the set E_λ defined by $E_\lambda = \{v : (A - \lambda I)v = 0\}$ is called the **eigenspace** or **characteristic space** of A associated with λ .

Remark:

- The eigenspace E_λ of A associated with a given λ is precisely the kernel or nullspace of the matrix $A - \lambda I$, i.e., $E_\lambda = \text{Ker}(A - \lambda I)$.
- E_λ is a subspace of V consisting of the zero vector and the eigenvectors of T corresponding to the eigenvalue λ .
- $(A - \lambda I)v = 0$ has a nonzero solution v if and only if $\det(A - \lambda I) = 0$.

Matrix Operations and Diagonalization

Definition 65 (algebraic multiplicity).

Let λ_i be an eigenvalue of a linear operator T (or matrix A). The **algebraic multiplicity** $\mu_T(\lambda_i)$ (or $\mu_A(\lambda_i)$) of the eigenvalue is its multiplicity as a root of the characteristic polynomial, i.e., the number of times $\lambda_i - \lambda$ occurs as a factor of $\det(T - \lambda I)$.

Definition 66 (geometric multiplicity).

The dimension of the eigenspace E_λ is referred to as the eigenvalue's **geometric multiplicity** $\gamma_T(\lambda)$, i.e., $\gamma_T(\lambda) = \dim(E_\lambda) = \dim(\text{Ker}(T - \lambda I))$.

Matrix Operations and Diagonalization

Example 67.

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix with $\lambda(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$.

- $\lambda \in \mathbb{C}$ is an eigenvalue of A if and only if there exists a nonzero eigenvector associated with λ .
- A is singular if and only if $\lambda_i = 0$ for some $i \in \{1, \dots, n\}$ (it has an eigenvalue that is equal to zero).
- The eigenvalues of a triangular matrix are equal to its diagonal entries.
- $\lambda(cI + A) = \{c + \lambda_1, c + \lambda_2, \dots, c + \lambda_n\}$.
- $\lambda(A^k) = \{\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k\}$.
- If A is nonsingular, then $\lambda(A^{-1}) = \{1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n\}$.
- $\lambda(A) = \lambda(A^\top)$.
- Similarity transformation: If S is nonsingular and $B = S^{-1}AS$, then $\lambda(A) = \lambda(B)$.

Matrix Operations and Diagonalization

Example 68.

Let T be a linear operator on a vector space V over the field $\mathbb{F}$, and let $g(t)$ be a polynomial with coefficients from $\mathbb{F}$. Prove that if x is an eigenvector of T with corresponding eigenvalue λ , then $g(T)(x) = g(\lambda)x$. That is, x is an eigenvector of $g(T)$ with corresponding eigenvalue $g(\lambda)$.

Matrix Operations and Diagonalization

- A linear operator T on a finite-dimensional vector space V is **diagonalizable** if there exists a (ordered) basis of V such that the matrix representation of T is a diagonal matrix.

Theorem 69.

*A linear operator T on a finite-dimensional vector space V is **diagonalizable** if and only if there exists a (ordered) basis of V consisting of eigenvectors of T . In other words, the sum of the dimensions of its eigenspaces is equal to $\dim(V)$.*

Definition 70 (Diagonalizable matrix).

A square matrix A is **diagonalizable** or non-defective if it is similar to a diagonal matrix. That is, if there exists an invertible matrix P (i.e. an element of the general linear group $GL_n(\mathbb{F})$) and a diagonal matrix D such that $P^{-1}AP = D$.

$$A \text{ diagonalizable} \iff \exists P \in GL_n(\mathbb{F}) \text{ s.t. } P^{-1}AP \text{ diagonal}$$

Or equivalently, $A = PDP^{-1}$. Note that such P and D are not unique.

Matrix Operations and Diagonalization

Theorem 71.

Let T be a linear operator on a finite-dimensional vector space V such that the characteristic polynomial of T splits. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . Then T is diagonalizable if and only if the algebraic multiplicity of λ_i is equal to $\dim(E_{\lambda_i})$ for all i .

Lemma 72.

A $n \times n$ matrix A is diagonalizable when the algebraic multiplicity equals the geometric multiplicity for each distinct eigenvalue.

Matrix Operations and Diagonalization

- A vector space V is called the **direct sum** of W_1 and W_2 if W_1 and W_2 are subspaces of V such that $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = V$. We denote that V is the direct sum of W_1 and W_2 by $V = W_1 \oplus W_2$.

Lemma 73.

A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if V is the direct sum of the eigenspaces of T . That is, $V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}$ providing $\lambda(T) = \{\lambda_1, \dots, \lambda_k\}$.

In other words, an endomorphism $T : V \rightarrow V$ is diagonalizable if and only if V is the direct sum of the eigenspaces of T .

Matrix Operations and Diagonalization

Lemma 74.

Let T be a linear operator on an n -dimensional vector space V . If T has n distinct eigenvalues, then T is diagonalizable.

Lemma 75.

An $n \times n$ matrix A is diagonalizable if it has n distinct eigenvalues, i.e. if its characteristic polynomial has n distinct roots; however, the converse may be false.

- The converse of above lemma is false. That is, it is not true that if T is diagonalizable, then it has n distinct eigenvalues. For example, the identity operator is diagonalizable even though it has only one eigenvalue.

Matrix Operations and Diagonalization

Example 76.

Let T be the linear operator on $P_2(\mathbb{R})$ defined by $T(f(x)) = f'(x)$ and let $\mathcal{B}$ be the standard ordered basis for $P_2(\mathbb{R})$. Show that T is not diagonalizable.

The matrix representation of T with respect to $\mathcal{B}$ is given by

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Therefore, the characteristic polynomial of T is given by $\det(A - tI) = -t^3$. Thus T has only one eigenvalue $\lambda = 0$ with algebraic multiplicity 3. Furthermore, $E_\lambda = N(T - \lambda I) = N(T)$, which implies E_λ contains constant polynomials and therefore $\dim(E_\lambda) = 1$. This results indicate that T is not diagonalizable (from theorem 71).

Matrix Operations and Diagonalization

Example 77.

Let T be the linear operator on $P_2(\mathbb{R})$ defined by $T(f(x)) = f(1) + f'(0)x + (f'(0) + f''(0))x^2$ and let $\mathcal{B}$ be the standard ordered basis for $P_2(\mathbb{R})$. Show that T is diagonalizable.

Matrix Operations and Diagonalization

Example 78.

Show that A given below is diagonalizable and compute A^n for any positive integer n .

$$A = \begin{pmatrix} 0 & -2 \\ 1 & 3 \end{pmatrix}$$

Matrix Operations and Diagonalization

Example 79.

For $A \in \mathbb{R}^{n \times n}$, define $e^A = \lim_{k \rightarrow \infty} B_k$, where $B_k = I + A + \frac{A^2}{2!} + \cdots + \frac{A^k}{k!}$. Show that

- Compute e^O and e^I , where O and I denote the zero and identity matrices, respectively.
- Suppose A is diagonalizable, i.e., $P^{-1}AP = D$. Show that $e^A = Pe^DP^{-1}$.

Matrix Operations and Diagonalization

Definition 80 (Spectral Radius).

The spectral radius of a matrix $A \in \mathbb{R}^{n \times n}$ is $\rho(A) = \max\{|\lambda_i|\}$, where $\lambda(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$, namely, the maximum of the magnitudes of the eigenvalues of A . Then $\rho : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ is a constant function of A .

Lemma 81.

Let A be a square matrix. We have $\lim_{k \rightarrow \infty} A^k = 0$ if and only if $\rho(A) < 1$.

Matrix Operations and Diagonalization

Why **the Cayley-Hamilton theorem**?

- The Cayley Hamilton Theorem is used to define concepts in control theory such as the controllability of linear systems.
- Other applications include to calculate the solutions of linear differential equations, the higher powers of the matrices, characteristic polynomials of the matrix, and the inverse of the matrix, ...

Matrix Operations and Diagonalization

Definition 82 (T -invariant).

Let V be a vector space and let $T : V \rightarrow V$ be linear. A subspace W of V is said to be **T -invariant (subspace of V)** if $T(x) \in W$ for every $x \in W$, that is, $T(W) \subseteq W$. If W is T -invariant, we define **the restriction of T on W** to be the function $T_W : W \rightarrow W$ defined by $T_W(x) = T(x)$ for all $x \in W$.

Example 83.

Suppose that T is a linear operator on a vector space V . Show that the following subspaces of V are T -invariant:

$$\{0\}, V, \mathcal{R}(T), \mathcal{N}(T), E_\lambda$$

Matrix Operations and Diagonalization

Definition 84 (T -cyclic subspace).

Let T be a linear operator on a vector space V , and let x be a nonzero vector in V . The subspace given by

$$W = \text{span}(\{x, T(x), T^2(x), \dots\})$$

is called the **T -cyclic subspace of V generated by x** .

- W is T -invariant.
- W is the "smallest" T -invariant subspace of V containing x .
- T -cyclic subspace is important for studying the *Cayley-Hamilton theorem* and the *matrix representations of nondiagonalizable linear operators*.

Matrix Operations and Diagonalization

Theorem 85.

Let T be a linear operator on a finite-dimensional vector space V , and let W be a T -invariant subspace of V . Then the characteristic polynomial of T_W divides the characteristic polynomial of T .

Sketchproof:

$$A = [T]_{\mathcal{B}_V} = \begin{pmatrix} B_1 & B_2 \\ O & B_3 \end{pmatrix}, \quad B_1 = [T_W]_{\mathcal{B}_W}$$

$$f(t) = \det(A - tI_n) = \det(B_1 - tI_k) \det(B_3 - tI_{n-k}) = g(t) \det(B_3 - tI_{n-k})$$

Matrix Operations and Diagonalization

Theorem 86.

Let T be a linear operator on a finite-dimensional vector space V and let W denote the T -cyclic subspace of V generated by a nonzero vector $v \in V$. Let $k = \dim(W)$. Then

- 1. $\mathcal{B} = \{v, T(v), T^2(v), \dots, T^{k-1}(v)\}$ is a basis for W .*
- 2. If $a_0v + a_1T(v) + \dots + a_{k-1}T^{k-1}(v) + T^k(v) = 0$, then the characteristic polynomial of T_W is $f(t) = (-1)^k(a_0 + a_1t + \dots + a_{k-1}t^{k-1} + t^k)$.*

Matrix Operations and Diagonalization

Theorem 87 (The Cayley-Hamilton theorem (operator)).

Let T be a linear operator on a finite-dimensional vector space V and let $f(t)$ be the characteristic polynomial of T . Then $f(T) = T_0$, the zero transformation. That is, T satisfies its characteristic equation.

Proof: We show that $f(T)(v) = 0$ for all $v \in V$. This is obvious if $v = 0$ because $f(T)$ is linear. Suppose that $v \neq 0$. Let W be the T -cyclic subspace generated by v , and suppose that $\dim(W) = k$. By Theorem 86, there exist scalars $a_0, \dots, a_{k-1}$ such that

$$a_0 v + a_1 T(v) + \dots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$$

with the characteristic polynomial of T_W given by $g(t) = (-1)^k(a_0 + a_1 t + \dots + a_{k-1} t^{k-1} + t^k)$. Combining these two equations yields

$$g(T)(v) = (-1)^k(a_0 I + a_1 T + \dots + a_{k-1} T^{k-1} + T^k)(v) = 0$$

Furthermore, by Theorem 85, $g(t)$ divides $f(t)$; hence there exists a polynomial $q(t)$ such that $f(t) = q(t)g(t)$. So $f(T)(v) = q(T)g(T)(v) = q(T)(g(T)(v)) = q(T)(0) = 0$.

Matrix Operations and Diagonalization

Theorem 88 (The Cayley-Hamilton theorem (matrix)).

Let A be an $n \times n$ matrix and let $f(\lambda)$ be the characteristic polynomial of A . Then $f(A) = O$, the $n \times n$ zero matrix. That is, if $f(\lambda) = c_0 + c_1\lambda^1 + \cdots + c_n\lambda^n$, then

$$f(A) = c_0 + c_1A^1 + \cdots + c_nA^n = O$$

Proof: Exercise. (Note that if $f(t) = \det(A - tI)$ is the characteristic polynomial of A , it is tempting to "prove" that $f(A) = O$ by saying " $f(A) = \det(A - AI) = \det(O) = 0$." But this argument is nonsense. Why?)

Matrix Operations and Diagonalization

- The advantage of a diagonalizable linear operator lies in the simplicity of its description. Such an operator has a diagonal matrix representation, or, equivalently, there is an ordered basis for the underlying vector space consisting of eigenvectors of the operator. However, **not every linear operator is diagonalizable**. $\Rightarrow$ Study alternative matrix representations for nondiagonalizable operators, i.e., jordan canonical form.

Matrix Operations and Diagonalization

Definition 89 (Jordan canonical form / Jordan normal form).

Let A be an $n \times n$ matrix such that the characteristic polynomial of A splits, then there exists a block diagonal matrix J such that A is similar to J (i.e., $A \sim J$), with J given by

$$J = \begin{bmatrix} J_1 & O & \cdots & O \\ O & J_2 & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & J_k \end{bmatrix}_{n \times n}, \text{ and } J_i := J_{s_i}(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & & \lambda_i & 1 \\ 0 & 0 & \cdots & 0 & \lambda_i \end{bmatrix}_{s_i \times s_i}$$

with $\sum_i s_i = n$. Here, J is called the **Jordan canonical form** of A and J_i is the Jordan block corresponds to λ_i .

Matrix Operations and Diagonalization

Remark:

- The definition in the previous page can also be written in a "linear operator" form. That is, every linear operator on a finite dimensional vector space whose characteristic polynomial splits has a Jordan canonical form that is unique up to the order of the Jordan blocks. Note that, it is **not** the case that the Jordan canonical form is completely determined by the characteristic polynomial of the operator. For example,

$$J = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \text{ and } J' = \begin{bmatrix} 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Matrix Operations and Diagonalization

Theorem 90.

Each Jordan block J_i is of the form $\lambda_i I + N$, where N is **nilpotent**.

- A matrix N is nilpotent if $N^k = 0$ for some $k \in \mathbb{N}$.

Corollary 91.

Suppose $t \in \mathbb{R}$ is a scalar. The exponential of Jordan block J_i of the form $e^{J_i t}$ can be expressed as the sum of finite terms, given by

$$e^{J_i t} = e^{(\lambda_i I + N)t} = e^{\lambda_i t} e^{Nt} = e^{\lambda_i t} \left[I + Nt + N^2 \frac{t^2}{2!} + \cdots + N^{k-1} \frac{t^{k-1}}{(k-1)!} \right]$$

provided N is nilpotent with $N^k = 0$.

Matrix Operations and Diagonalization

Remark:

- Suppose $t, \sigma \in \mathbb{R}$ is a scalar, then $e^{A(t+\sigma)} = e^{At}e^{A\sigma}$.
- In general, given two distinct matrices $A, B \in \mathbb{R}^{n \times n}$, $e^{A+B} \neq e^A e^B$. Since

$$e^{A+B} = I + (A+B) + \frac{1}{2!}(A^2 + AB + BA + B^2) + \dots$$

$$e^A e^B = (I + A + \frac{1}{2!}A^2 + \dots)(I + B + \frac{1}{2!}B^2 + \dots) = I + (A+B) + \frac{1}{2!}(A^2 + 2AB + B^2) + \dots$$

Thus, $e^{A+B} = e^A e^B$ if $AB = BA$, i.e., $[A, B] = 0$. If a pair of matrices (A, B) satisfies $[A, B] = 0$, we say A and B **commutes**.

- **Baker-Campbell-Hausdorff Formula:** Let A and B be square matrices. Then $e^A e^B = e^C$, with C given by

$$C = A + B + \frac{1}{2}[A, B] + \frac{1}{12}\left\{ [[B, A], A] + [[B, A], B] \right\} + \dots$$

Matrix Operations and Diagonalization

Example 92.

Show that

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 \\ 0 & \lambda_i & 1 & & 0 \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & 0 & & \lambda_i & 1 \\ 0 & 0 & \cdots & 0 & \lambda_i \end{bmatrix}_{s_i \times s_i} \Rightarrow e^{J_i t} = e^{\lambda_i t} \begin{bmatrix} 1 & t & \frac{t^2}{2!} & \cdots & \frac{t^{s_i-1}}{(s_i-1)!} \\ 0 & 1 & t & & \vdots \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & & & 1 & t \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

Matrix Operations and Diagonalization

Definition 93.

Let T be a linear operator on a vector space V and let λ be a scalar. A nonzero vector x in V is called a **generalized eigenvector of T corresponding to λ** if $(T - \lambda I)^p(x) = 0$ for some positive integer p .

Definition 94.

Let T be a linear operator on a vector space V , and let λ be an eigenvalue of T . The **generalized eigenspace of T corresponding to λ** , denoted by K_λ , is the subset of V defined by

$$K_\lambda = \{x \in V : (T - \lambda I)^p(x) = 0 \text{ for some } p \in \mathbb{N}\}$$

Note that K_λ consists of the zero vector and all generalized eigenvectors corresponding to λ .

Matrix Operations and Diagonalization

Definition 95.

Let T be a linear operator on a vector space V and let x be a generalized eigenvector of T corresponding to λ . Suppose p is the smallest positive integer for which $(T - \lambda I)^p(x) = 0$. Then the ordered set

$$\left\{ (T - \lambda I)^{p-1}(x), (T - \lambda I)^{p-2}(x), \dots, (T - \lambda I)(x), x \right\}$$

is called a **cycle of generalized eigenvectors of T corresponding to λ** . The vectors $(T - \lambda I)^{p-1}(x)$ and x are called the initial vector and the end vector of the cycle, respectively. We say that the length of the cycle is p .

Matrix Operations and Diagonalization

- Let T be a linear operator on a finite-dimensional vector space V , and suppose that the characteristic polynomial of T splits. Recall in Lemma 73 that T is *diagonalizable* if and only if V is the direct sum of the eigenspaces of T . If T is diagonalizable, then the eigenspaces and the generalized eigenspaces coincide. The next result extends Lemma 73 to the nondiagonalizable case.

Lemma 96.

Let T be a linear operator on a finite-dimensional vector space V whose characteristic polynomial splits. Then V is the direct sum of the generalized eigenspaces of T .

Matrix Operations and Diagonalization

Example 97.

Find the Jordan canonical form of $A = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 0 & 5 \\ -1 & -1 & 4 \end{pmatrix}$

Matrix Operations and Diagonalization

Example 98.

Let T be the linear operator on $P_2(\mathbb{R})$ defined by $T(g(x)) = -g(x) - g'(x)$. Find a Jordan canonical form of T and a Jordan canonical basis for T .

Matrix Operations and Diagonalization

Remark: Computation of Matrix Exponential

Given an $n \times n$ (constant) matrix A , then the matrix exponential e^{At} can be computed by:

1. Diagonalization
2. Cayley Hamilton Theorem
3. Laplace Transform

Matrix Operations and Diagonalization

- The Laplace transform of $x(t)$ is defined by

$$\hat{x}(s) = \mathcal{L}(x(t)) = \int_0^{\infty} e^{-st} x(t) dt,$$

if the integral exists, where s is a complex number frequency parameter. Also, recall the derivative property of the Laplace transform:

$$\mathcal{L}(\dot{x}(t)) = \int_0^{\infty} e^{-st} \dot{x}(t) dt = x e^{-st} \Big|_0^{\infty} - \int_0^{\infty} -s e^{-st} x(t) dt = s\hat{x}(s) - x(0).$$

Theorem 99 (Laplace Transform Method).

Let $\dot{x} = Ax$. Taking the Laplace transform results in $s\hat{x}(s) - x(0) = A\hat{x}(s)$, or $(sI - A)\hat{x}(s) = x(0)$, so we have $\hat{x}(s) = (sI - A)^{-1}x(0)$. We find the solution by inverting the transform: $x(t) = \mathcal{L}^{-1}[(sI - A)^{-1}]x(0)$.

Matrix Operations and Diagonalization

Example 100.

Let t be a scalar, find e^{At} for the following matrices:

- $A = \begin{pmatrix} 5 & -2 & 4 \\ 0 & 1 & 0 \\ -2 & 2 & -1 \end{pmatrix}$

- $A = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$

- $A = \begin{pmatrix} 0 & \omega \\ \omega & 0 \end{pmatrix}$

Inner Product Spaces

- Inner Product / Norm
- Orthogonality / Orthonormal Basis
- The Adjoint of a Linear Operator
- Normal / Self-Adjoint Operators
- Unitary / Orthogonal Operators
- Orthogonal Projections / the Spectral Theorem
- The Singular Value Decomposition / the Pseudoinverse

Inner Product Spaces

- Many geometric notions such as angle, length, and perpendicularity in $\mathbb{R}^2$ and $\mathbb{R}^3$ may be extended to more general real and complex vector spaces. All of these ideas are related to the concept of inner product.

Inner Product Spaces

Definition 101 (Inner Product).

Let X be a vector space over $\mathbb{F}$. An **inner product** on X is a mapping $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{F}$ such that $\forall x, y, z \in X, \alpha \in \mathbb{F}$, we have

1. $\langle x, x \rangle \geq 0$ (positivity) and $\langle x, x \rangle = 0 \Leftrightarrow x = 0$ (definiteness)
2. $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$ (additivity in the first component)
3. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ (homogeneity in the first component)
4. $\langle x, y \rangle = \overline{\langle y, x \rangle}$ (conjugate symmetry)

A vector space X over $\mathbb{F}$ endowed with a specific inner product $\langle \cdot, \cdot \rangle$ is called an **inner product space**, denoted by $(X, \langle \cdot, \cdot \rangle)$.

Example 102.

Verify the inner product is **antilinear** or **conjugate linear** in the second component, i.e.,

$$\langle x, \alpha y + \beta z \rangle = \bar{\alpha} \langle x, y \rangle + \bar{\beta} \langle x, z \rangle$$

Inner Product Spaces

Example 103.

- If $X = \mathbb{F}^n$, the (standard) inner product of two vectors $x, y \in \mathbb{F}^n$ is defined by $\langle x, y \rangle = y^* x = \sum_{i=1}^n x_i \bar{y}_i$.
 - If $X = \mathbb{F}^{n \times n}$, the inner product of two matrices $A, B \in \mathbb{F}^{n \times n}$ is defined by $\langle A, B \rangle = \text{tr}(B^* A)$, where B^* denotes the **conjugate transpose** or adjoint of B , i.e., $(B^*)_{ij} = \overline{B_{ji}}$. This inner product is called the **Frobenius inner product**.
 - If $X = C(\Omega)$, the inner product of two continuous complex-valued functions $f, g \in C(\Omega)$ is defined by $\langle f, g \rangle = \int_{\Omega} f(t) \overline{g(t)} dt$.
 - If $X = C^m(\Omega)$, the inner product of two m -tuple functions $f, g \in C^m(\Omega)$ is defined by $\langle f, g \rangle = \int_{\Omega} g^*(t) f(t) dt$.
-
- An **Hilbert space** $\mathcal{H}$ is a complete inner product space.

Inner Product Spaces

Remark:

- Two distinct inner products on a given vector space yield two distinct inner product spaces. For instance, it can be shown that both

$$\langle f, g \rangle_1 = \int_0^1 f(t)g(t)dt, \quad \langle f, g \rangle_2 = \int_{-1}^1 f(t)g(t)dt$$

are inner products on the vector space $P(\mathbb{R})$. Even though the underlying vector space is the same, however, these two inner products yield two different inner product spaces. For example, the polynomials $f(t) = t$ and $g(t) = t^2$ are orthogonal in the second inner product space, but not in the first.

Inner Product Spaces

Definition 104 (Norm).

A mapping $\| \cdot \| : X \rightarrow \mathbb{F}$ on a vector space X over a field $\mathbb{F}$ is called a **norm** if $\forall x, y \in X$ and $\alpha \in \mathbb{F}$, it satisfies

1. $\|x\| \geq 0$ (non-negativity) and $\|x\| = 0 \Leftrightarrow x = 0$ (definiteness)
2. $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)
3. $\|\alpha x\| = |\alpha| \cdot \|x\|$ (homogeneity)

If X is an inner product space, then $\|x\| = \sqrt{\langle x, x \rangle}$ is the **induced norm**.

Inner Product Spaces

Example 105.

- The ℓ_1 norm on $\mathbb{R}^n$ is $\|x\|_1 = \sum_{i=1}^n |x_i|$.
- The **Euclidean norm** (or ℓ_2 norm) on $\mathbb{R}^n$ is $\|x\|_2 = \sqrt{\langle x, x \rangle} = (\sum_{i=1}^n x_i^2)^{1/2}$.
- The **supremum norm** (or ℓ_∞ norm) on $\mathbb{R}^n$ is $\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$.
- The p -norm of $\mathbb{R}^n$ is $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ for $p \geq 1$.
- The **matrix norm** (or **induced norm** or **operator norm**) on $\mathbb{R}^{n \times n}$ is

$$\|A\| = \max_{x \in \mathbb{R}^n, \|x\|=1} \|Ax\| = \max_{x \in \mathbb{R}^n} \frac{\|Ax\|}{\|x\|}.$$

- The **Frobenius norm** on $\mathbb{R}^{n \times n}$ is

$$\|A\|_F = \sqrt{\text{tr}(A^* A)}$$

Inner Product Spaces

Theorem 106 (Equivalent norms).

*Let X be a (normed) vector space. Any two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on X are **equivalent** if there exist positive constants C_1 and C_2 such that for all $x \in V$,*

$$C_1\|x\|_a \leq \|x\|_b \leq C_2\|x\|_a$$

Remark:

- Equivalent norms define equivalent metrics and hence the same topology and the same Cauchy sequences.
- All norms on a finite-dimensional vector space (over $\mathbb{R}$ or $\mathbb{C}$) are equivalent.

Inner Product Spaces

Theorem 107 (Equivalence of norms on $\mathbb{R}^n$).

For any two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on $\mathbb{R}^n$, there exist positive constants α and β such that for all $x \in \mathbb{R}^n$,

$$\alpha\|x\|_a \leq \|x\|_b \leq \beta\|x\|_a$$

- e.g.: $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\|x\|_\infty$

Remark:

Any norms on $\mathbb{R}^n$ induce the same topology, i.e., the same collection of open subsets, the same set of convergence sequences, and so on. Equivalent norms define the same notions of continuity and convergence, and for many purposes do not need to be distinguished. The uniform structure defined by equivalent norms on a vector space is a **uniform isomorphism**.

Inner Product Spaces

Example 108 (Parallelogram Law).

Let X be an inner product space with induced norm defined by $\|x\| = \sqrt{\langle x, x \rangle}$ for $x \in X$. Then for $x, y \in X$,

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

Proof: From the definition of norm, we have

$$\begin{aligned}\|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y \rangle + \langle x - y, x - y \rangle \\ &= \left(\langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \right) + \left(\langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle \right) \\ &= 2\langle x, x \rangle + 2\langle y, y \rangle\end{aligned}$$

Inner Product Spaces

Theorem 109 (Cauchy-Schwarz inequality).

Let X be a real inner product space. For all $x, y \in X$,

$$|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle = \|x\|^2 \|y\|^2.$$

Equality holds if and only if $x = ky$ for $k \in \mathbb{R}$ or $y = 0$.

Proof: The case $y = 0$ is trivial. Assume $y \neq 0$. For all scalars k , we have

$$\begin{aligned} 0 \leq \|x - ky\|^2 &= \langle x - ky, x - ky \rangle = \langle x, x \rangle - \langle x, ky \rangle - \langle ky, x \rangle + \langle ky, ky \rangle \\ &= \|x\|^2 - k\langle x, y \rangle - k\langle y, x \rangle + k^2\|y\|^2. \end{aligned}$$

In particular, taking $k = \frac{\langle x, y \rangle}{\langle y, y \rangle}$ yields $0 \leq \langle x, x \rangle - \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle}$ and so $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$. The equality holds if and only if $\|x - ky\|^2 = 0$, i.e., $x = ky$.

Inner Product Spaces

Example 110.

Let X be an inner product space. The inner product space $\langle \cdot, \cdot \rangle$ is a continuous map from $X \times X \rightarrow \mathbb{F}$. That is, if $x_1 \rightarrow x_2$ and $y_1 \rightarrow y_2$ in X , then $\langle x_1, y_1 \rangle \rightarrow \langle x_2, y_2 \rangle$.

Proof: For all $x_i, y_i \in X$, the Cauchy-Schwarz inequality implies that

$$\begin{aligned} |\langle x_1, y_1 \rangle - \langle x_2, y_2 \rangle| &= |\langle x_1 - x_2, y_1 \rangle + \langle x_2, y_1 - y_2 \rangle| \\ &\leq |\langle x_1 - x_2, y_1 \rangle| + |\langle x_2, y_1 - y_2 \rangle| \\ &\leq |x_1 - x_2| |y_1| + |x_2| |y_1 - y_2| \rightarrow 0 \end{aligned}$$

which leads to the continuity of the inner product.

Inner Product Spaces

Definition 111 (Orthogonality).

Let X be an inner product space. $x, y \in X$ are said to be **orthogonal** if $\langle x, y \rangle = 0$ ($x \perp y$). $x \in X$ is said to be **orthogonal** to a set S if $\langle x, s \rangle = 0$ ($x \perp S$) for all $s \in S$.

Definition 112 (Orthonormal Basis).

An orthonormal basis $\mathcal{B}_V = \{u_i\}_{i=1}^n$ for a finite-dimensional vector space V is a collection of vectors for which

$$\langle u_i, u_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}, \quad 1 \leq i, j \leq n$$

If $v \in V$, then there exists a unique expression $v = \sum_{i=1}^n a_i u_i$ for some $\{a_i\}_{i=1}^n$, where $a_i = \langle v, u_i \rangle$. The a_i are sometimes called **Fourier coefficients**.

Inner Product Spaces

Remark:

- Any finite-dimensional inner product space has an orthonormal basis, which may be obtained from an arbitrary basis using the Gram-Schmidt process.
- The above statement is not always true for infinite-dimensional space. That is, an infinite-dimensional inner product space may not have an orthonormal basis.
- The infinite-dimensional inner product space has an orthonormal basis if the space is **complete** (every Cauchy sequence in the space converge to some point in that space) and **separable** (contain a countable set which is dense in that space). For example, a separable Hilbert space $\mathcal{H}$.

Inner Product Spaces

- In an infinite-dimensional setting, we study the "orthonormal system" instead and construct an orthonormal basis accordingly.
- Since not all infinite-dimensional vector space has an orthonormal basis, we focus on the "Hilbert space" setting.

Definition 113 (Orthonormal System).

A system of vectors $\{\psi_i\}_{i=1}^{\infty}$ for an infinite-dimensional vector space $\mathcal{H}$ is an orthonormal system if $\langle \psi_i, \psi_j \rangle = \delta_{ij}$ for all $i, j \in \mathbb{N}$.

An orthonormal system $\{\psi_i\}_{i=1}^{\infty}$ is called an orthonormal basis for $\mathcal{H}$ if for every $v \in \mathcal{H}$, we have $v = \sum_k \alpha_k \psi_k$ for some $\alpha_k \in \mathbb{C}$. The $\alpha_k = \langle v, \psi_k \rangle$ are called **Fourier coefficients**.

- It's important to note that for the above definition to make sense, we have to concern with the convergence of the infinite sum of the series.

Inner Product Spaces

Theorem 114.

Let $\{\psi_i\}_{i=1}^{\infty}$ be an orthonormal system in $\mathcal{H}$. The following statements are equivalent.

1. $\{\psi_i\}_{i=1}^{\infty}$ is an orthonormal basis for $\mathcal{H}$.
2. If $\langle x, \psi_k \rangle = 0$ for all k , then $x = 0$.
3. $\text{span}\{\psi_i\}$ is **dense** in $\mathcal{H}$, i.e., every vector in $\mathcal{H}$ is a limit of a sequence of vector in $\text{span}\{\psi_i\}$.
4. For every $x \in \mathcal{H}$, $\|x\|^2 = \sum_k |\langle x, \psi_k \rangle|^2$

Inner Product Spaces

Example 115 (Find the matrix representation of L given bases for V and W).

Let $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$ and let $\mathcal{B}_V = \{u_1, \dots, u_n\}$ and $\mathcal{B}_W = \{v_1, \dots, v_m\}$ be orthonormal bases of V and W , respectively. Let $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear. Find the matrix representation of L given bases for V and W .

Inner Product Spaces

Example 116.

Let V be an inner product space and $S = \{v_1, \dots, v_n\}$ be an orthogonal subset of V consisting of nonzero vectors. If $y \in \text{span}(S)$, then

$$y = \sum_{i=1}^n \frac{\langle y, v_i \rangle}{\|v_i\|^2} v_i.$$

Since $y \in \text{span}(S)$, we can write $y = \sum_{i=1}^n a_i v_i$. Then for all $1 \leq j \leq n$, we have

$$\langle y, v_j \rangle = \left\langle \sum_{i=1}^n a_i v_i, v_j \right\rangle = a_j \langle v_j, v_j \rangle$$

which implies $a_i = \frac{\langle y, v_i \rangle}{\|v_i\|^2}$.

- What if V is infinite-dimensional?

Inner Product Spaces

Example 117.

Let $\mathcal{H} = (C([0, 2\pi]), \langle \cdot, \cdot \rangle)$ be an inner product space where $C([0, 2\pi])$ is the space of continuous complex-valued functions defined on the interval $[0, 2\pi]$ and the inner product defined by

$$\langle f, g \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(t) \bar{g}(t) dt$$

Verify the basis $\mathcal{B} = \{f_n : f_n(t) = e^{int}, n \in \mathbb{N}\}$ is an orthonormal basis.

Note, every element $f \in \mathcal{H}$ can be expressed as $f = \sum_n c_n f_n$ where

$$c_n = \langle f, f_n \rangle = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-int} dt$$

is the n -th **Fourier coefficient** for the continuous function $f \in \mathcal{H}$.

Inner Product Spaces

Definition 118 (Orthogonal complement).

Let S be a nonempty subset of an inner product space X . We define $S^\perp$ (read "S perp") to be the set of all vectors in X that are orthogonal to every vector in S ; that is

$$S^\perp = \{x \in X : \langle x, s \rangle = 0 \ \forall s \in S\}$$

The set S is called the **orthogonal complement** of S .

Inner Product Spaces

Definition 119 (Gram-Schmidt process).

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and $S = \{w_1, \dots, w_n\}$ be a linearly independent subset of V . Define $S' = \{v_1, \dots, v_n\}$, where $v_1 = w_1$ and

$$v_k = w_k - \sum_{j=1}^{k-1} \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} v_j, \quad \text{for } 2 \leq k \leq n$$

Then S' is an orthogonal set of nonzero vectors such that $\text{span}(S') = \text{span}(S)$. If each vector is further normalized, then we get an orthonormal basis for V .

- $\{1, x, x^2, \dots\}$ for $P(\mathbb{R})$ with $\langle f, g \rangle = \int_{-1}^1 f(t)g(t)dt \Rightarrow$ **Legendre polynomials**.
- $\{1, x, x^2, \dots\}$ for $P(\mathbb{R})$ with $\langle f, g \rangle = \int_{-1}^1 \frac{f(t)g(t)}{\sqrt{1-t^2}}dt \Rightarrow$ **Chebyshev polynomials**.

Inner Product Spaces

Example 120.

Suppose that $S = \{v_1, \dots, v_k\}$ is an orthonormal set in an n -dimensional inner product space V . Then by applying the Gram Schmidt process to an extend basis $\{v_1, \dots, v_k, w_{k+1}, \dots, w_n\}$, we get an orthonormal basis $\{v_1, \dots, v_k, v_{k+1}, \dots, v_n\}$ for V .

1. If $W = \text{span}(S)$, then $S' = \{v_{k+1}, \dots, v_n\}$ is an orthonormal basis for $W^\perp$.
2. If W is any subspace of V , then $\dim(V) = \dim(W) + \dim(W^\perp)$.

Inner Product Spaces

Example 121 (Parseval's Identity).

Let $\{v_1, \dots, v_n\}$ be an orthonormal basis for V . For any $x, y \in V$, show that

$$\langle x, y \rangle = \sum_{i=1}^n \langle x, v_i \rangle \overline{\langle y, v_i \rangle}$$

Inner Product Spaces

- Recall in previous discussion, we defined the conjugate transpose A^* of a matrix A . For a linear operator T on an inner product space V , we can define a related linear operator on V called the adjoint of T , whose matrix representation with respect to any orthonormal basis for V is $[T]^*$.
- Although many of the following results hold under different mathematical settings (e.g. finite-dimensional, bounded operator,...), we will focus on discussing the infinite dimensional counterpart to the adjoint of a matrix, namely the adjoint of a bounded linear operator.

Inner Product Spaces

- Let $(X, \langle \cdot, \cdot \rangle_X)$ and $(Y, \langle \cdot, \cdot \rangle_Y)$ be two inner product spaces. The linear operator $L : X \rightarrow Y$ is **bounded** if

$$\sup_{\|x\| \leq 1} \|Lx\| < \infty.$$

The space of all bounded linear operators from X to Y is denoted by $\mathcal{B}(X, Y)$.

Inner Product Spaces

Theorem 122 (Representation theorem).

Let $(X, \langle \cdot, \cdot \rangle_X)$ be an inner product space (over $\mathbb{F}$) and let $g : X \rightarrow \mathbb{F}$ be a bounded linear functional on X . Then there exists a unique vector $y \in X$ such that $g(x) = \langle x, y \rangle$ for all $x \in X$.

Intuition:

- Suppose $L \in \mathcal{B}(X, Y)$. For each $y \in Y$, the functional $f_y(x) = \langle Lx, y \rangle_Y$ is a bounded linear functional on X . Hence, the Riesz representation theorem guarantees the existence of a unique $y^* \in X$ such that for all $x \in X$,

$$\langle Lx, y \rangle_Y = f_y(x) = \langle x, y^* \rangle_X$$

This gives rise to an operator $L^* : Y \rightarrow X$ defined by $L^*y = y^*$.

Inner Product Spaces

Definition 123 (Adjoint operator).

Let $(X, \langle \cdot, \cdot \rangle_X)$ and $(Y, \langle \cdot, \cdot \rangle_Y)$ be two inner product spaces and let $L : X \rightarrow Y$ be a linear operator. Then the **adjoint operator** $L^* : Y \rightarrow X$ of L is defined by $\langle Lx, y \rangle_Y = \langle x, L^*y \rangle_X$ for all $x \in X$ and $y \in Y$. If such adjoint operator exists, then L^* is linear and a unique operator of L .

- For an infinite-dimensional inner product space, the adjoint of a linear operator L may not exist. Although the uniqueness and linearity of L^* still holds, the existence of the adjoint is not guaranteed (see Exercise 125).
- Every bounded linear operator between Hilbert spaces has an (unique) adjoint.
- If λ is an eigenvalue of L , then $\bar{\lambda}$ is the eigenvalue of L^* .

Inner Product Spaces

Example 124.

Let X, Y, Z be inner product spaces and let $L, U : X \rightarrow Y$ and $T : Y \rightarrow Z$ with the adjoints L^*, U^* and T^* exist, show that

1. $(L + U)^* = L^* + U^*$
2. $(cL)^* = \bar{c}L^*$
3. $L^{**} := (L^*)^* = L$
4. $(TL)^* = L^*T^*$

Inner Product Spaces

Example 125.

Let V be the vector space of all sequences $\sigma = (\sigma(1), \dots, \sigma(n), \dots)$ such that $\sigma(i) \neq 0 \in \mathbb{R}$ for only finitely many positive integers i and let $\langle \sigma, \mu \rangle = \sum_{i=1}^{\infty} \sigma(i)\mu(i)$. Define $T : V \rightarrow V$ by $T(\sigma)(k) = \sum_{i=k}^{\infty} \sigma(i)$ for every positive integer k . (Notice that the series σ and the infinite series in T both converge because $\sigma(i) \neq 0$ for only finitely many i .)

1. Prove that T is a linear operator on V .
2. For each positive integer n , let e_n be the sequence defined by $e_n(k) = \delta_{n,k}$. Prove that $\{e_1, e_2, \dots\}$ is an orthonormal basis for V .
3. Let $\sigma_n = e_1 + e_n$ and $W = \text{span}\{\sigma_n : n > 2\}$. Prove that $e_1 \notin W$, so $W \neq V$.
4. Prove that $W^\perp = 0$, and conclude that $W \neq (W^\perp)^\perp$.
5. Prove that for any positive integer n , $T(e_n) = \sum_{i=1}^n e_i$.
6. Prove that T has no adjoint.
7. Show that the space V is not complete.

Inner Product Spaces

Definition 126 (Self-Adjoint operator).

Let $(X, \langle \cdot, \cdot \rangle_X)$ be an inner product space and let $L : X \rightarrow X$ be a linear operator on X . We say L is **self-adjoint** if $L^* = L$.

- Every eigenvalue of a self-adjoint operator L is real.

Definition 127 (Normal operator).

Let $(X, \langle \cdot, \cdot \rangle_X)$ be an inner product space and let $L : X \rightarrow X$ be a linear operator on X . We say L is **normal** if $LL^* = L^*L$.

- Self-adjoint $\Rightarrow$ Normal, but Normal $\nRightarrow$ Self-adjoint.

Inner Product Spaces

Remark:

- An $n \times n$ real or complex matrix A is self-adjoint (Hermitian) if $A = A^*$. An $n \times n$ real or complex matrix A is normal if $AA^* = A^*A$.
- Suppose that A is a *real skew-symmetric matrix*, i.e., $A^T = -A$. Then A is normal.
- Let A be an $n \times n$ Hermitian (self-adjoint) matrix. Then there is a unitary matrix P such that $P^{-1}AP$ is diagonal (A is unitarily equivalent to a diagonal matrix). If A is a real symmetric matrix, there is a real orthogonal matrix P such that $P^{-1}AP$ is diagonal.

Inner Product Spaces

Example 128.

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space and let T be a normal operator on V . Then the following statements are true.

1. $\|T(x)\| = \|T^*(x)\|$ for all $x \in V$.
2. $T - cI$ is normal for every $c \in \mathbb{F}$.
3. If x is an eigenvector of T , then x is also an eigenvector of T^* . In fact, if $T(x) = \lambda x$, then $T^*(x) = \bar{\lambda}x$.
4. If λ_1 and λ_2 are distinct eigenvalues of T with corresponding eigenvectors x_1 and x_2 , i.e., $x_1 \in E_{\lambda_1}(T)$ and $x_2 \in E_{\lambda_2}(T)$, then x_1 and x_2 are orthogonal.
5. If, in addition, T is self-adjoint, then all eigenvalues of T are real.

Inner Product Spaces

Theorem 129.

- *Let T be a linear operator on a finite-dimensional complex inner product space V . Then T is normal if and only if there exists an orthonormal basis for V consisting of eigenvectors of T .*
- *Let T be a linear operator on a finite-dimensional real inner product space V . Then T is self-adjoint if and only if there exists an orthonormal basis $\mathcal{O}$ for V consisting of eigenvectors of T .*

Proof Idea: Applying Schur's Theorem (there exists an orthonormal basis for V such that the matrix representation of T is upper triangular)

Inner Product Spaces

Example 130.

A linear operator T on a finite-dimensional inner product space is called **positive definite** [**positive semidefinite**] if T is self-adjoint and $\langle T(x), x \rangle > 0$ [$\langle T(x), x \rangle \geq 0$] for all $x \neq 0$. Show that

1. T is positive definite [semidefinite] if and only if all of its eigenvalues are positive [nonnegative].
2. If T is positive semidefinite, then $A = B^*B$ for some square matrix B , where A is the matrix representation of T .

Inner Product Spaces

Let A be an $n \times n$ symmetric matrix, then $\forall x \in \mathbb{R}^n$ with $x \neq 0$,

- $A \succ 0$ is **positive definite** $\Leftrightarrow x^T A x > 0 \Leftrightarrow \lambda_i > 0 \Leftrightarrow \Delta_k > 0, k = 1, \dots, n$
- $A \prec 0$ is **negative definite** $\Leftrightarrow x^T A x < 0 \Leftrightarrow \lambda_i < 0 \Leftrightarrow (-1)^k \Delta_k > 0, k = 1, \dots, n$
- $A \succeq 0$ is **positive semidefinite** $\Leftrightarrow x^T A x \geq 0 \Leftrightarrow \lambda_i \geq 0$

Here, Δ_k is the k -th principal minor of A for $1 \leq k \leq n$ and the condition related to Δ_k is referred to as Sylvester's criterion.

Remark!!

- $\Delta_k > 0, \forall k = 1, \dots, n-1, \Delta_n = 0 \Rightarrow A \succeq 0$
- $(-1)^k \Delta_k > 0, \forall k = 1, \dots, n-1, \Delta_n = 0 \Rightarrow A \preceq 0$
- $\Delta_k \geq 0, \forall k = 1, \dots, n \not\Rightarrow A \succeq 0$

- e.g.: $A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0.5 \end{pmatrix}, x = \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$

Inner Product Spaces

Example 131.

An $n \times n$ real matrix A is said to be a **Gramian matrix** if there exists a real (square) matrix B such that $A = B^T B$. Prove that A is a Gramian matrix if and only if A is symmetric and all of its eigenvalues are nonnegative.

Inner Product Spaces

Definition 132 (Unitary / Orthogonal operator).

Let T be a linear operator on a finite-dimensional inner product space V (over $\mathbb{F}$). If $\|T(x)\| = \|x\|$ for all $x \in V$, we call T a **unitary operator** if $\mathbb{F} = \mathbb{C}$ and an **orthogonal operator** if $\mathbb{F} = \mathbb{R}$.

- In the infinite-dimensional case, an operator satisfying the preceding norm requirement is generally called an **isometry**.
- If V and W are two normed linear spaces over the same field, then they are **isometrically isomorphic** iff there exists a bijective linear isometry $T : V \rightarrow W$.
- Any two infinite dimensional separable Hilbert spaces (over $\mathbb{C}$) are linearly isometric.
- Any n -dimensional Hilbert space (over $\mathbb{C}$) is linear isometric to $\mathbb{C}^n$.
- Any infinite dimensional separable Hilbert space is linearly isometric to ℓ^2 .

Inner Product Spaces

Remark:

- A square matrix A is called an orthogonal matrix if $A^\top A = AA^\top = I$ and unitary if $A^* A = AA^* = I$.
- Let A and B be complex $n \times n$ matrices. We say that B **is unitarily equivalent to** A if there is an $n \times n$ unitary matrix P such that $B = P^* A P = P^{-1} A P$. We say that B **is orthogonally equivalent to** A if there is an $n \times n$ orthogonal matrix P such that $B = P^\top A P = P^{-1} A P$.

Inner Product Spaces

Example 133.

Let T be a linear operator on a finite-dimensional inner product space V . Then the following statements are equivalent.

1. $TT^* = T^*T = I$.
2. $\langle T(x), T(y) \rangle = \langle x, y \rangle$ for all $x, y \in V$. (T preserves inner products)
3. If $\mathcal{B}_V$ is an orthonormal basis for V , then $T(\mathcal{B}_V)$ is an orthonormal basis for V .
4. $\|T(x)\| = \|x\|$ for all $x \in V$

- An isomorphism of V onto W is a vector space isomorphism T of V onto W which also preserves inner products.

Inner Product Spaces

Example 134.

Let A be an $n \times n$ symmetric matrix, then

- $\lambda(A) = \{\lambda_1, \dots, \lambda_n\}$, λ_i is real.
- $A = PDP^\top$, where P is orthogonal and $D = \text{diag}(\lambda_1, \dots, \lambda_n)$.
- A has a set of n mutually orthogonal, real, and nonzero eigenvectors $v_1, \dots, v_n$.
- $A = \sum_{i=1}^n \lambda_i v_i v_i^\top$, v_i orthonormal.

Inner Product Spaces

Remark:

If X , Y are finite dimensional linear spaces and $\mathcal{R}(L)$ is finite dimensional, then

$$X = \mathcal{N}(L) \oplus \mathcal{N}(L)^\perp ; Y = \mathcal{R}(L) \oplus \mathcal{R}(L)^\perp$$

Inner Product Spaces

Theorem 135.

Suppose that X, Y are inner product spaces and that $L : X \rightarrow Y$. Then

$$\mathcal{N}(L) = \mathcal{R}(L^*)^\perp ; \mathcal{N}(L^*) = \mathcal{R}(L)^\perp$$

If $\mathcal{R}(L)$ is finite dimensional, then

$$\mathcal{N}(L)^\perp = \mathcal{R}(L^*) ; \mathcal{N}(L^*)^\perp = \mathcal{R}(L)$$

(Otherwise, $\mathcal{N}(L)^\perp = \overline{\mathcal{R}(L^)}$; $\mathcal{N}(L^*)^\perp = \overline{\mathcal{R}(L)}$.)*

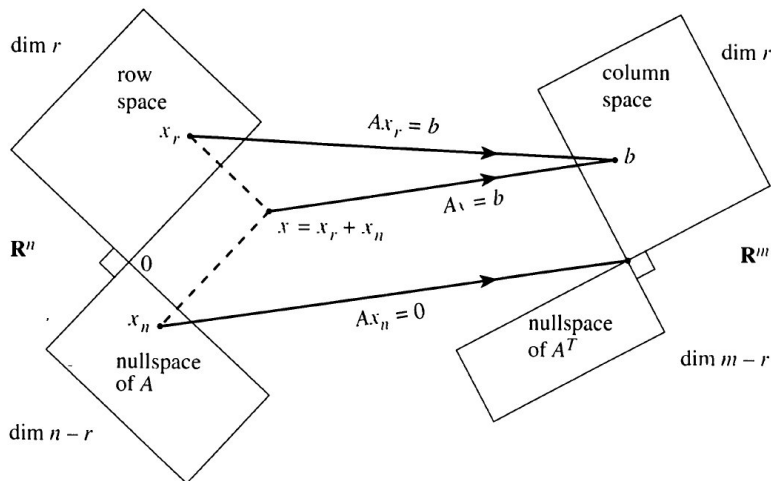
Theorem 136.

Suppose that X, Y are inner product spaces and that $L \in \mathcal{B}(X, Y)$. If $\mathcal{R}(L)$ is finite dimensional, then

$$X = \mathcal{N}(L) \oplus \mathcal{R}(L^*) ; Y = \mathcal{R}(L) \oplus \mathcal{N}(L^*)$$

Inner Product Spaces

Remark: **Four fundamental subspaces of a matrix (finite-dimensional case)**



Inner Product Spaces

Theorem 137.

Suppose that X, Y are inner product spaces and that $L \in \mathcal{B}(X, Y)$. If $\mathcal{R}(L)$ is finite dimensional, then

$$\mathcal{N}(L) = \mathcal{N}(L^*L) ; \mathcal{R}(L^*) = \mathcal{R}(L^*L)$$

$$\mathcal{N}(L^*) = \mathcal{N}(LL^*) ; \mathcal{R}(L) = \mathcal{R}(LL^*)$$

(Otherwise, $\overline{\mathcal{R}(L^)} = \overline{\mathcal{R}(L^*L)}$ and $\overline{\mathcal{R}(L)} = \overline{\mathcal{R}(LL^*)}$.)*

Inner Product Spaces

Example 138.

Let T be a linear operator on a finite-dimensional inner product space V . Show that

1. $\mathcal{N}(T^*T) = \mathcal{N}(T)$. Deduce that $\text{rank}(T^*T) = \text{rank}(T)$.
2. $\text{rank}(T) = \text{rank}(T^*)$. Deduce from 1 that $\text{rank}(TT^*) = \text{rank}(T)$.

Inner Product Spaces

- Why adding closure for the following: $(M^\perp)^\perp = \overline{M}$?

Example 139.

The orthogonal complement of a subset of a Hilbert space is a closed linear subspace.

Inner Product Spaces

Definition 140 (orthogonal projection).

Let V be an inner product space and let $T : V \rightarrow V$ be a projection on V . We say that T is an **orthogonal projection** if $\mathcal{R}(T)^\perp = \mathcal{N}(T)$ and $\mathcal{N}(T)^\perp = \mathcal{R}(T)$.

Theorem 141.

Let V be an inner product space and let T be a linear operator on V . Then T is an orthogonal projection if and only if T has an adjoint T^ and $T^2 = T = T^*$.*

Inner Product Spaces

Theorem 142 (The Spectral Theorem of Linear Operator).

Suppose that T is a linear operator on a finite-dimensional inner product space V over $\mathbb{F}$ with the distinct eigenvalues $\lambda_1, \dots, \lambda_k$. Assume that T is normal if $\mathbb{F} = \mathbb{C}$ and that T is self-adjoint if $\mathbb{F} = \mathbb{R}$. For each $1 \leq i \leq k$, let W_i be the eigenspace of T corresponding to the eigenvalue λ_i , and let T_i be the orthogonal projection of V on W_i . Then the following statements are true.

1. $V = W_1 \oplus W_2 \oplus \dots \oplus W_k$.
2. If W'_i denotes the direct sum of the subspaces W_j for $j \neq i$, then $W_i^\perp = W'_i$.
3. $T_i T_j = \delta_{ij} T_i$ for $1 \leq i, j \leq k$.
4. $I = T_1 + T_2 + \dots + T_k$ (the resolution of the identity operator induced by T).
5. $T = \lambda_1 T_1 + \lambda_2 T_2 + \dots + \lambda_k T_k$ (the spectral decomposition of T).

*The set $\{\lambda_1, \dots, \lambda_k\}$ of eigenvalues of T is called the **spectrum** of T .*

Inner Product Spaces

Theorem 143 (Singular Value Theorem for Linear Transformations).

Let V and W be finite-dimensional inner product spaces and let $T : V \rightarrow W$ be a linear transformation of rank r . Then there exist orthonormal bases $\{v_1, \dots, v_n\}$ for V and $\{u_1, \dots, u_m\}$ for W and positive scalars $\sigma_1 \geq \dots \geq \sigma_r > 0$ such that

$$T(v_i) = \begin{cases} \sigma_i u_i, & \text{if } 1 \leq i \leq r \\ 0, & \text{if } i > r \end{cases}$$

*Moreover, the σ_i are uniquely determined by T , and are called the **singular values** of T .*

Inner Product Spaces

Uniqueness:

$$\langle T^*(u_i), v_j \rangle = \langle u_i, T(v_j) \rangle = \begin{cases} \sigma_i, & \text{if } i = j \leq r \\ 0, & \text{if } i > r \end{cases}$$

and hence for any $1 \leq i \leq m$,

$$T^*(u_i) = \sum_{j=1}^n \langle T^*(u_i), v_j \rangle v_j = \begin{cases} \sigma_i v_i, & \text{if } i = j \leq r \\ 0, & \text{if } i > r \end{cases}$$

For $i \leq r$,

$$T^* T(v_i) = T^*(\sigma_i u_i) = \sigma_i T^*(u_i) = \sigma_i^2 v_i$$

Therefore each v_i is an eigenvector of $T^* T$ with corresponding eigenvalue σ_i^2 if $i \leq r$ and 0 if $i > r$.

Inner Product Spaces

Remark:

- Let A be an $m \times n$ matrix of rank r with the positive singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$, and let Σ be the $m \times n$ matrix defined by

$$\Sigma_{ij} = \begin{cases} \sigma_{ij}, & \text{if } i = j \leq r \\ 0, & \text{otherwise} \end{cases}$$

Then there exists an $m \times m$ unitary matrix U and a $n \times n$ unitary matrix V such that $A = U\Sigma V^*$. Such factorization of A is called a singular value decomposition of A .

Inner Product Spaces

Example 144.

If $A = U\Sigma V^\top$ is a SVD of $A \in \mathbb{R}^{m \times n}$ with $\sigma_1 \geq \dots \geq \sigma_r > \sigma_{r+1} = \dots = \sigma_p = 0$ with $p = \min(m, n)$, we have the following statements:

1. $\text{rank}(A) = r$.
2. $\mathcal{N}(A) = \text{span}\{v_{r+1}, \dots, v_n\}$; $\mathcal{R}(A) = \text{span}\{u_1, \dots, u_r\}$.
3. $A = \sum_{j=1}^r \sigma_j u_j v_j^\top = U_r \Sigma_r V_r^\top$ (decompose A into sum of rank-1 matrices).
4. $\|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2}$; $\|A\|_2 = \sigma_1$.
5. $\sigma_j = \sqrt{\lambda_j(A^\top A)}$, $j = 1, \dots, p$, where $\lambda_j(A^\top A)$ is the j th largest eigenvalue of $A^\top A$.
6. v_i and u_i are orthonormalized eigenvectors of $A^\top A$ and $AA^\top$, respectively.
7. v_i are the right singular vectors of A and u_i are the left singular vectors of A .

Inner Product Spaces

Why **Pseudoinverse**?

- It is desirable to have a linear transformation from W to V that captures some of the essence of an inverse of $T : V \rightarrow W$ even if T is not invertible. A simple approach to this problem is to focus on the "part" of T that is invertible, namely, the restriction of T to $\mathcal{N}(T)^\perp$. The idea is to define an invertible linear transformation $L : \mathcal{N}(T)^\perp \rightarrow \mathcal{R}(T)$ by $L(x) = T(x)$ for all $x \in \mathcal{N}(T)^\perp$ and use the inverse of L to construct a linear transformation from W to V that salvages some of the benefits of an inverse of T .

Inner Product Spaces

Definition 145 (Pseudoinverse / Moore-Penrose generalized inverse).

Let V and W be finite-dimensional inner product spaces over the same field, and let $T : V \rightarrow W$ be a linear transformation. Let $L : \mathcal{N}(T)^\perp \rightarrow \mathcal{R}(T)$ be the linear transformation defined by $L(x) = T(x)$ for all $x \in \mathcal{N}(T)^\perp$. The **pseudoinverse** (or Moore-Penrose generalized inverse) of T , denoted by $T^\dagger$, is defined as the unique linear transformation from W to V such that

$$T^\dagger(y) = \begin{cases} L^{-1}(y), & \text{for } y \in \mathcal{R}(T) \\ 0, & \text{for } y \in \mathcal{R}(T)^\perp \end{cases}$$

The pseudoinverse of a linear transformation T on a finite-dimensional inner product space exists even if T is not invertible. Furthermore, if T is invertible, then $T^\dagger = T^{-1}$ because $\mathcal{N}(T)^\perp = V$, and L coincides with T .

Inner Product Spaces

Lemma 146.

Suppose that V and W be finite-dimensional inner product spaces over the same field and let $T : V \rightarrow W$ be a linear transformation of rank r . Let $\{v_1, v_2, \dots, v_n\}$ and $\{u_1, u_2, \dots, u_m\}$ be orthonormal bases for V and W , respectively, and let $\sigma_1 \geq \dots \geq \sigma_r > 0$ be the nonzero singular values of T satisfying $T(v_i) = \sigma_i u_i$ for $1 \leq i \leq r$ and 0 otherwise. Then $\{v_1, v_2, \dots, v_r\}$ is a basis for $\mathcal{N}(T)^\perp$, $\{v_{r+1}, \dots, v_n\}$ is a basis for $\mathcal{N}(T)$, $\{u_1, u_2, \dots, u_r\}$ is a basis for $\mathcal{R}(T)$, and $\{u_{r+1}, \dots, u_m\}$ is a basis for $\mathcal{R}(T)^\perp$. Let L be the restriction of T to $\mathcal{N}(T)^\perp$, as in the definition of pseudoinverse. Then $L^{-1}(u_i) = \frac{1}{\sigma_i} v_i$ for $1 \leq i \leq r$. Therefore,

$$T^\dagger(u_i) = \begin{cases} \frac{1}{\sigma_i} v_i, & \text{if } 1 \leq i \leq r \\ 0, & \text{if } r < i \leq m \end{cases}$$

Inner Product Spaces

Remark:

- Let A be an $m \times n$ matrix of rank r with a singular value decomposition $A = U\Sigma V^*$ and positive singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. Let $\Sigma^\dagger$ be the $m \times n$ matrix defined by

$$\Sigma_{ij}^\dagger = \begin{cases} \frac{1}{\sigma_{ij}}, & \text{if } i = j \leq r \\ 0, & \text{otherwise} \end{cases}$$

then $A^\dagger = V\Sigma^\dagger U^*$ is a singular value decomposition of $A^\dagger$.

Inner Product Spaces

Example 147.

Let V and W be finite-dimensional inner product spaces, and let $T : V \rightarrow W$ be linear. Show that

1. $TT^\dagger T = T$.
2. $T^\dagger TT^\dagger = T^\dagger$.
3. Both $T^\dagger T$ and $TT^\dagger$ are self-adjoint.
4. If T is one-to-one, then $T^\dagger T$ is invertible and $T^\dagger = (T^* T)^{-1} T^*$.
5. If T is onto, then $TT^\dagger$ is invertible and $T^\dagger = T^* (TT^*)^{-1}$.
6. $T^\dagger T$ is the orthogonal projection of V on $\mathcal{N}(T)^\perp$.
7. $TT^\dagger$ is the orthogonal projection of W on $\mathcal{R}(T)$.

Statements 1 to 3 are called the Penrose conditions, and they characterize the pseudoinverse of a linear transformation, i.e., let $U : W \rightarrow V$ be linear transformations such that $TUT = T$, $UTU = U$, and both UT and TU are self-adjoint, then $U = T^\dagger$.

Inner Product Spaces

Theorem 148.

Consider the system of linear equations $Ax = b$. where $A \in \mathbb{R}^{m \times n}$ is a $m \times n$ matrix and $b \in \mathbb{R}^m$. If $z = A^\dagger b$, then z has the following properties.

- If $Ax = b$ is consistent (underdetermined $m < n$), then z is the unique solution to the system having minimum norm. That is, z is a solution to the system, and if y is any solution to the system, then $\|z\| \leq \|y\|$ with equality if and only if $z = y$.
- If $Ax = b$ is inconsistent (overdetermined $m > n$), then z is the unique best approximation to a solution having minimum norm. That is, $\|Az - b\| \leq \|Ay - b\|$ for any $y \in \mathbb{R}^n$, with equality if and only if $Az = Ay$. Furthermore, if $Az = Ay$, then $\|z\| \leq \|w\|$ with equality if and only if $z = y$.

Inner Product Spaces

Euclidean norm ($\|\cdot\|$) vs Generalized H-norm ($\|\cdot\|_H$)

Consider $Ax = b$, where $A \in \mathbb{R}^{n \times m}$, $x \in \mathbb{R}^m$, $b \in \mathbb{R}^n$. Assume $H \succ 0 \in \mathbb{R}^{n \times n}$.

	Inconsistent linear system $n > m$ (Curve fitting)	underdetermined linear system $n < m$ (Minimum norm)
$\ \cdot\ $	x^* minimizes $\ b - Ax\ ^2$ $\iff A^T Ax^* = A^T b$ $\implies x^* = (A^T A)^{-1} A^T b$	x^* minimizes $\ x\ $, subject to $Ax = b$ $\iff x^* \in \mathcal{R}(A^T)$ $\implies x^* = A^T (AA^T)^{-1} b$
$\ \cdot\ _H$	x^* minimizes $\ b - Ax\ _H^2$ $\iff A^T H A x^* = A^T H b$ $\implies x^* = (A^T H A)^{-1} A^T H b$	x^* minimizes $\ x\ _H$, subject to $Ax = b$ $\iff x^* \in \mathcal{R}(H^{-1} A^T)$ $\implies x^* = H^{-1} A^T (A H^{-1} A^T)^{-1} b$

The End